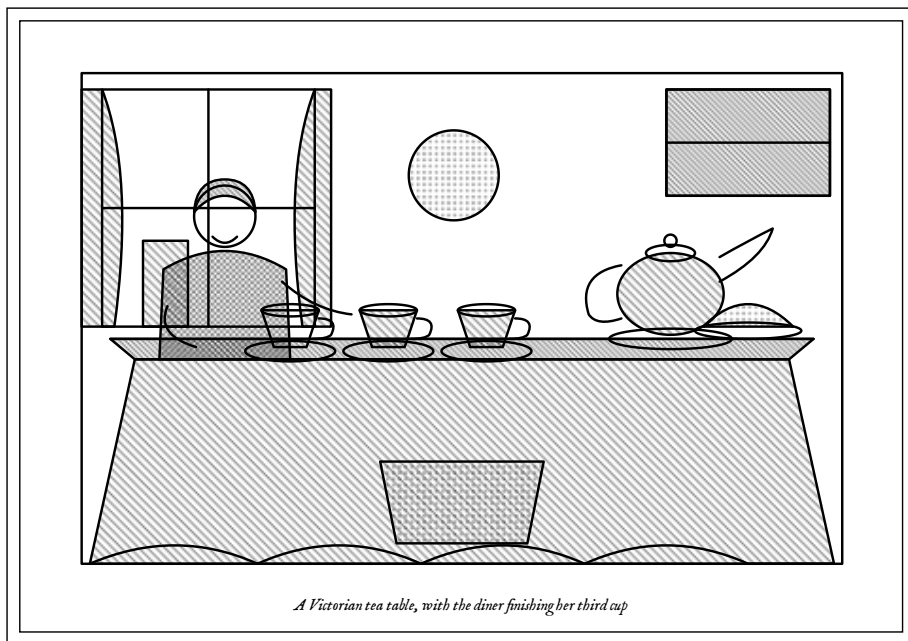




CHAPTER III

Utility and Diminishing Marginal Utility

Why does the same cup of tea please a little less each time?



This chapter explains how economists describe satisfaction, how they count it with an imaginary unit, and why each extra unit of a good adds less than the one before. A tea table and a student buying exercise books show the rule at work, and the chapter ends with the limits of the rule.

AN ELEMENTARY TREATISE ON MICROECONOMICS



*Being a First Course in Demand, Supply, Cost and the Behaviour of
Markets*

S. SHIR, M. ROUD

2.6 A production table shows 0 cars with 50 buses, 5 cars with 40 buses, and 10 cars with 20 buses. Find the opportunity cost of a car in each step and say what happens to it.

2.7 Give one reason why a real production possibility curve bows outward, and one event that would shift it outward.

Before moving on. The production possibility curve shows the largest combinations of two goods. Points on it use all resources; points inside waste some; points outside are out of reach. Its slope shows opportunity cost, and growth moves the whole curve outward.

* * *

WORKED EXAMPLES: CHANGING THE SETTING

The first example is Rafiq above, with clean numbers. The next three change the setting. Each uses the order Given, Find, Formula, Steps, Answer, Check.

Given: a farmer has 3 acres. On one acre, she can grow 2,500 kg of rice or 1,500 kg of jute. *Find:* the opportunity cost of 1 kg of jute, in rice. *Formula:* quantity given up ÷ quantity gained. *Steps.* One acre of jute gives up 2,500 kg of rice and gains 1,500 kg of jute. So 1 kg of jute costs $2,500 \div 1,500 = 1.67$ kg of rice. *Answer:* about 1.67 kg of rice. *Check:* the reverse cost of 1 kg of rice is $1,500 \div 2,500 = 0.6$ kg of jute, and 1.67×0.6 is close to 1, as it should be.

Given: in one hour, a bakery can make 6 dozen rolls or 3 dozen cakes. *Find:* the opportunity cost of one cake, in rolls. *Steps.* Convert to single items: 6 dozen = 72 rolls and 3 dozen = 36 cakes. One hour of cakes gives up 72 rolls and gains 36 cakes, so one cake costs $72 \div 36 = 2$ rolls. *Answer:* 2 rolls. *Check:* working in dozens gives $6 \div 3 = 2$ as well. The slip to avoid is to compare dozens of one good with single items of the other.

Given: a workshop can produce these combinations with all resources fully used.

COMBINATIONS OF CHAIRS AND TABLES A WORKSHOP CAN MAKE

CHAIRS	TABLES
0	30
10	26
20	18
30	0

Find: the opportunity cost of an extra chair at each step, and whether 20 chairs with 20 tables is attainable. *Steps.* From 0 to 10 chairs, tables fall from 30 to 26, so 4 tables are given up for 10 chairs: 0.4 tables per chair. From 10 to 20 chairs, 8 tables are given up: 0.8 per chair. From 20 to 30 chairs, 18 tables are given up: 1.8 per chair. At 20 chairs the most tables possible is 18, so 20 tables is out of reach. *Answer:* the cost rises at each step, and the point is unattainable. *Check:* the sum of the three losses is $4 + 8 + 18 = 30$, the full number of tables.

* * *

SUMMARY

Before moving on: the curve shows the attainable, and growth moves it.

Four examples, each a little less tidy than the last.

Rung 2: a farm with less tidy numbers gives a ratio that is not whole.

Rung 3: the units change from dozens to single items.

Rung 4: a table combines costs, points inside and outside, and a rising ratio.

The table is read row by row, and the cost is found between rows.

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First edition

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Case file, illustrative. Rafiq has 40 hours a week for two jobs. In one hour he can shelve 20 books or pack 10 online parcels. If he spends all 40 hours on shelving, he completes 800 books and 0 parcels. If he spends all 40 hours on packing, he completes 400 parcels and 0 books.

Given: the record above. *Find:* the opportunity cost of one extra parcel, in books. *Formula:* opportunity cost per unit = quantity given up ÷ quantity gained. *Steps.* One hour of packing gains 10 parcels. The same hour, spent on shelving, would have produced 20 books. So 10 parcels cost 20 books, and one parcel costs $20 \div 10 = 2$ books. *Answer:* each extra parcel gives up 2 books. *Check:* going from 800 books to 0 books gives up 800 books, and $800 \div 400$ parcels is also 2 books for each parcel. The two methods agree.

The opposite choice has the reverse cost. One extra book gives up $10 \div 20 = 0.5$ of a parcel. In general the opportunity cost of A in terms of B is the inverse of the opportunity cost of B in terms of A. A common slip is to forget this, and to give 2 books where 0.5 of a parcel is wanted. Always say which good is being measured, and in which unit.

Case file, illustrative. Nurul Meghna, who founded Meghna and Sons in 1996, has Tk 500,000 to spend on one of two uses: new titles for the shop, or an online site. The opportunity cost of choosing the new titles is the online site, and the opportunity cost of choosing the online site is the new titles.

Before moving on. Opportunity cost is the best alternative given up. When outputs are known, divide the quantity given up by the quantity gained. Rafiq's extra parcel costs 2 books. Nurul's choice of one use of Tk 500,000 costs the other.

* * *

Recall: economics defined, Chapter 1, section 1.1 'The Meaning and Scope of Economics'.

The Steps for Rafiq follow the usual order: Given, Find, Formula, Steps, Answer, Check.

The opportunity cost of books is the reverse ratio.

Opportunity cost can also be a choice between two uses of money.

Before moving on: the best alternative given up, measured in the units of the other good.

2.3 THE PRODUCTION POSSIBILITY CURVE

DRAWING THE CHOICES

Rafiq's choice is not limited to all shelving or all packing. He may share his 40 hours in any proportion. A picture can show every such choice together. Before drawing it, test one more case from his record. Suppose he gives 20 hours to shelving and 20 hours to packing. Shelving gives $20 \times 20 = 400$ books. Packing gives $20 \times 10 = 200$ parcels. That is a third combination: 400 books and 200 parcels.

Plot the combinations and join them. The result is a **production possibility curve**, which means a line showing the largest combinations of two goods that can be produced with all resources used fully. Like the demand schedule in Chapter 4, it holds other things the same: Rafiq's hours, his speed and his tools stay fixed while the choices change. Each point on the line is one way of dividing the 40 hours.

To read the figure, look at the axes. The horizontal axis shows

A picture can show every choice open to Rafiq at once.

Recall: laws and the same-conditions rule, Chapter 1, section 1.4 'Economic Laws'.

Production possibility curve. A line showing the largest combinations of two goods that can be produced with all resources used fully.

The production possibility curve joins the largest combinations of two goods.

The axes show books and

hours, and other wants stand ready to take its place. What has no end is the whole list.

Before moving on. A want is something a person would like to have, whatever the price. One want may be met for a time, but the list as a whole keeps growing. This is half of the picture. The other half follows.

SCARCITY AND RESOURCES

Think of a market stall. The owner has a small table, a few hours and a purse of money. The customers wish for many goods, but the stall can offer only what the table can hold, what the hours allow and what the purse can buy. The owner's wishes are larger than the stall. The same is true for every household, firm and country, for the same reason: the things used to make goods and services are limited.

Economists call these things resources. Resources are the inputs used to make goods and services: land and raw materials, the work of people, tools and machines, money, and time. A bakery's resources include its oven, its flour, its bakers and its hours. A country's resources include its farmland, its rivers, its factories and its workers. Not one of them is unlimited at a given time.

When unlimited wants meet limited resources, the result is **scarcity**. Scarcity means the condition that resources are limited while wants are unlimited, so choices must be made. A family cannot buy everything on its list, and a country cannot build everything its citizens wish for. Scarcity is the reason economics exists. If every want could be met at once, nothing would need to be chosen and nothing would need to be studied.

Scarcity has two common misreadings. The first is that scarcity means rarity. Air is common, but a pure breath of it in a crowded city may be scarce, and the same is true of clean water. The second is that scarcity belongs only to poor people. A billionaire also has limited time and must choose. Scarcity affects rich and poor, and it applies to every society.¹

Scarcity forces choice. Faced with limited resources, a person, a firm or a government must decide which wants to meet and which to leave. A choice may be made carefully or carelessly, openly or without thought. Doing nothing is also a choice, because the hours and the money go somewhere. The question is never whether to choose, but how well.

Each decision maker meets scarcity from a different side. A household has a fixed income and must divide it among food, rent, school fees and savings. A firm has a limited budget, a limited number of workers and a limited number of working hours, and must decide what to make and how much. A government has limited tax income and must divide it among roads, schools, hospitals and defence. The three decision makers differ in size, yet all three must choose, and every choice leaves some wants unmet.

Before moving on. Resources are limited; wants are not. Scarcity means with alternative uses is the common feature of every economic problem.

Before moving on: a want is a wish, and the whole list of wants has no end.

GO SLOWLY HERE.

A market stall shows the limit that meets unlimited wishes.

Resources are the things used to produce goods and services.

Scarcity. The condition that resources are limited while wants are unlimited, so choices must be made.

Scarcity is the gap between unlimited wants and limited resources.

Scarcity is not the same as rarity, and it is not the same as poverty.

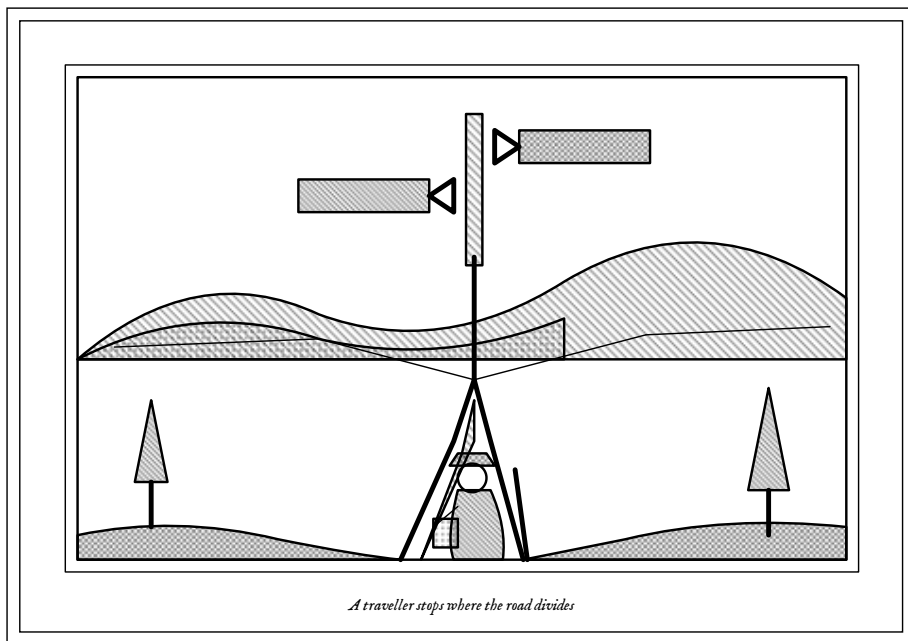
Choice. Word in use. In daily life, the thing selected, or the freedom to select. In economics, picking one use of limited resources and so giving up the others.

Choice is the act of selecting among uses of limited resources.

Households, firms and governments each face scarcity in their own way.

Before moving on: scarcity is

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CHAPTER 11

Scarcity, Choice and Opportunity Cost

Every road taken leaves another road behind.



This chapter shows why every choice gives something up, how to measure what is given up, and how a single curve can picture the choices open to a person, a firm or a country.

THINK IT THROUGH

1.T1 In a town, the price of notebooks rose in March, and notebook sales fell. In the same month, the school gave a free notebook to every student. A newspaper says the price rise alone caused the fall. Using the words *ceteris paribus*, explain why this claim is weak.

1.T2 A friend says, "Economic laws are useless, because they are not exact." Using Section 1.4, write a reply of three sentences.

WHERE PEOPLE GO WRONG

Many people believe that economics is the study of money. Money appears in many economic questions, but it is a tool, not the subject. The subject is choice under limited resources, and choices are made even where no money changes hands, as when a student chooses how to spend a free afternoon.

Some students think that micro means small things and macro means large things. This is a common confusion. The words refer to the unit studied. A large firm with thousands of staff is still one firm, so questions about it are micro. A question about the total number of jobs in the country is macro, however the jobs are spread.

Another confusion is that positive statements are true and normative statements are worthless. Positive statements can be false, as the case of 39,000 copies showed. Normative statements are not worthless: choices about what ought to be are the reason many people study economics. Facts alone cannot settle them by facts alone.

The last confusion concerns *ceteris paribus*. Some students read it as a claim that nothing else ever changes. In fact, the phrase tells us to imagine other things unchanged, so that one cause can be studied on its own. Without it, an economist cannot tell whether a fall in sales came from a higher price or from some other change.

IN HISTORY

Alfred Marshall (1842 to 1924) was an English economist who taught at the University of Cambridge. He began to write his book *Principles of Economics* in 1881 and published it in 1890. For many years it was the leading economics textbook in England. In it, Marshall described economics as the study of people in "the ordinary business of life".

The phrase matters for two reasons. First, it makes people, not money, the centre of the subject. Second, it treats economics as part of ordinary life. Buying a notebook, hiring an assistant and choosing a price are all examples of that business. The choices of Meghna and Sons are of exactly this kind, as are the choices of the father and the cloth seller in our opening puzzle.

Marshall's wording is still quoted in textbooks today. This chapter's definition adds one idea that his phrase does not state directly: the focus on limited resources, which explains why choice

Many people think economics is the study of money, but it studies choice.

Micro does not mean small things, and macro does not mean large things.

A positive Statement is not always true, and a normative Statement is not worthless.

Ceteris paribus is a thinking device, not a claim about the world.

Alfred Marshall's 1890 book gave economics a definition that centres on people.

The definition moves the subject from money towards everyday human conduct.

The definition still serves as a starting point, with later additions.

This chapter needs no earlier knowledge.

A market square shows many people facing one problem: choice.

The seller also faces limits, so both sides of the stall must choose.

The two choices look different but share one shape.

This puzzle is the subject of the book; the chapter plan follows.

GO SLOWLY HERE.

Begin with your own week: wants that cost more than your money.

Three plain words carry the idea: want, good and resource.

Economics is the study of

You will need: nothing. The chapter starts from things you already do: spending, saving, buying and selling. Each subject word is explained at the moment it first appears, in plain English, with a small example.

Picture a market square on a Saturday morning in a town of the 1880s. Stalls sell rice, cloth, fruit, pens and notebooks. Hundreds of people walk between them, and each carries a short list of things to buy and a limited amount of money. One father wants shoes for his daughter and a new lamp for the house. He can pay for only one of the two today. Which does he choose, and how does he decide?

Now consider the other side of the stalls. A cloth seller has one table, ten hours of daylight and a fixed amount of money to spend on stock. She can buy more cotton or more silk for tomorrow, but she cannot buy as much of both as she would like. Whatever she chooses, she gives up something else. She must also decide the price at which to sell. A price that is too high leaves her cloth unsold, and a price that is too low leaves her with too little money.

The father's choice is about a lamp, and the seller's choice is about silk. They look unrelated, yet they share one shape. Wants are many, the means to satisfy them are few, and so someone must choose. A harder question follows. When thousands of such choices are made each day, why does the square not fall into disorder? Why do shoes, lamps and cloth appear at the stalls in roughly the quantities that people wish to buy, at prices they can pay?

No single person in the square plans all of this, and that fact makes the puzzle worth studying. This book builds its answer step by step, and it begins with the plainest question of all: what does this subject study? Section 1.1 gives the meaning of the subject. Section 1.2 divides it into two branches. Section 1.3 separates statements of fact from statements of opinion. Section 1.4 explains the kind of law that economists can state, and the condition that every such law carries.

* * *

1.1 THE MEANING AND SCOPE OF ECONOMICS

Begin with something you know well. Imagine a student who receives Tk 100 for the week. She wants a notebook at Tk 60, a meal in a café at Tk 60, and a bus ticket at Tk 30. Together these wants cost Tk 150, which is more than Tk 100. She cannot have all three, so she must choose, and each choice means giving up something else. This small case holds the whole subject in one picture.

Three plain words are needed. A want is a wish to have something, such as a notebook. A good is anything that satisfies a want. Some books also say commodity or product; this book always says good, and uses service only for help given by a person, such as a lesson. A resource is anything used to make goods: land, people's work, tools and money. Keep these three words in mind, because the next definition uses all of them.

The student's problem has a name in the subject. **Economics**

1.4 ECONOMIC LAWS

Water boils at 100 degrees Celsius at sea level, and a stone released from the hand falls to the ground. These statements hold in every place, without exception. Now consider this one: when the price of a good rises, consumers usually buy less of it. Is that equally certain? Many students find this step confusing, because both statements are called laws. The difference is the subject of this section.

An **economic law** means a statement of how people usually behave, true when other conditions stay the same. Take the statement about price. When the price of a notebook rises, most consumers buy fewer notebooks. The word usually is part of the law. Some consumers may still buy the same number, but the usual pattern is clear enough to guide a seller or a government.

The condition has a name. **Ceteris paribus** is a Latin phrase meaning all other things stay the same.² Economists use it to study one change at a time. To test how the price of notebooks affects purchases, we imagine that incomes, tastes and the prices of other goods stay fixed. Otherwise we could not tell which change caused the result. The phrase is a method of thinking. It does not claim that other things never change; in real life they change all the time.

IN SHORT

An economic law is a usual pattern, and it holds *ceteris paribus*: all other things staying the same.

How do economic laws differ from natural laws? A law of nature, such as the fall of a stone, describes things that do not choose, so it can be exact and the same everywhere. An economic law describes people, who have different tastes and change their minds. For this reason, an economic law states a tendency, not a certainty. People also cannot be placed in a laboratory and held still, so economists must use the words *ceteris paribus* to separate one cause from many.

Because resources are limited, every economy faces the same three basic problems. It must decide what to produce, which means which goods, and how many of each. It must decide how to produce, which means with which resources and methods. It must decide for whom to produce, which means how the goods will be shared among people. The market square in our opening puzzle answers these questions every day without any one person planning the answers.

Before moving on. An economic law states how people usually behave, and it holds *ceteris paribus*. It is a tendency, not a certainty, because people choose. Every economy must decide what, how and for whom to produce. You should now be able to say why the words *ceteris paribus* must accompany an economic law.

WORKED CASE: SORTING QUESTIONS ABOUT

² Latin was the language of ancient Rome. Scholars in Europe used it for many centuries, so many technical phrases still come from it.

GO SLOWLY HERE.

A law of nature is exact, so what can an economic law claim?

An economic law states a usual pattern of behaviour under a condition.

Economic law. A statement of how people usually behave, true when other conditions stay the same.

Ceteris paribus names the condition that every economic law carries.

Ceteris paribus. A Latin phrase meaning all other things stay the same, used to study one change at a time.

Economic laws are less exact than natural laws because people choose.

Every economy faces three basic problems.

Recap of the section in a few lines.

it studies people, who differ from one another, change their minds and cannot be placed in a laboratory. One exception to the comparison with physics follows from this: economic results are less exact.

Before moving on. Economics is the study of choice. It exists because wants are many and resources are limited. It covers consumption, production, exchange and distribution, and it is both a science, which explains, and an art, which advises. You should now be able to say, in one sentence, why a student with Tk 100 and wants worth Tk 150 faces an economic problem.

1.2 MICRO AND MACRO

Start with maps. A street map of Dhanmondi shows each lane, shop and house, and it helps you find one bookshop. A map of the whole country does not show the lanes, but it shows the roads that join the cities, and it helps you plan a long journey. Neither map is wrong, and neither replaces the other. They are two ways of looking at the same land. Economics uses the same two views, and each has its own name.

The close view is called **microeconomics**. Microeconomics means the study of the choices made by individual consumers and individual firms. A consumer is a person who buys a good to use it, not to sell it again. The word begins with the Greek word for small.¹ In microeconomics we ask how one consumer decides what to buy, how one firm decides how much to make and at what price, and how the choices of many consumers and firms meet in a market.

Case file, illustrative. Meghna and Sons is a micro unit. In 2025 its sales were Tk 14,400,000, its profit was Tk 750,000, and it had 7 staff. A microeconomist who studies this shop asks how its owner chose the price of its books, how many staff to employ, and how many copies to keep in stock. Each of these questions concerns one firm.

The wide view is called **macroeconomics**. Macroeconomics means the study of the whole economy, including output, prices and employment. Output is the total amount of goods made in a country in a period of time. The price level is the general level of prices across the country. Employment is the number of people who have paid work. A macroeconomist does not ask about one shop. The questions are of this kind: why did the price level of the whole country rise, or why did total employment fall?

Recap of the section in a few lines.

GO SLOWLY HERE.

A street map and a country map show different views of one land.

Microeconomics is the close view of single consumers and firms.

Microeconomics. The branch of economics that studies choices made by individual consumers and firms.

Market. Word in use. In daily life, a place with stalls where people buy and sell. In economics, any arrangement in which buyers and sellers deal with each other, in one place or by telephone or screen.

A microeconomist asks questions about one firm, such as this shop.

Macroeconomics is the wide view of the whole economy.

Macroeconomics. The branch of economics that studies the whole economy, including output, prices and employment.

Economy. Word in use. In daily life, a saving, as in an economy of effort. In economics, the whole system by which the people of a country make, share and use goods.

¹ *Mikros* is Greek for small and *makros* is Greek for large. Many scientific words are built from Greek parts in this way.

Every consumer has an income, which is the money received in a period such as a month. Wants are many, but income is limited. A taka spent on one good cannot be spent again on another. So every act of consumption is also a choice: to buy one good is to decide not to buy something else with the same money.

How should a consumer choose? Chapter 3 showed that each extra unit of a good adds less utility than the one before it. This fact guides the choice. Suppose a student has Tk 100 left at the end of a month. The next Tk 20 spent on tea adds more utility than the next Tk 20 spent on a fourth magazine. A sensible student moves money from the magazine to the tea. The student stops when no further move would add utility.

The rule has limits. Real consumers do not calculate every purchase. Habit, advice from friends and the mood of the day all play a part. The idea of a sensible consumer describes the usual case, and it is not a picture of every person. It is still useful, because it explains why the amounts that consumers spend on different goods change when prices change.

Before moving on. Consumption is the use of goods and services to satisfy wants, and it happens when the good is used. Consumers have limited income, so they spend it where an extra taka adds the most utility. The price a consumer pays does not show the satisfaction received.

* * *

6.2 CONSUMERS' SURPLUS

Return to the woman with the pot. What is the source of her extra Tk 40? To answer, we first need a name for the limit she carried in her mind. Many students find this step confusing, because the limit is not written anywhere. It exists only as a decision that the consumer would make if the price rose.

Willingness to pay means the highest price a buyer would pay for one more unit of a good. For the woman, the willingness to pay for the pot was Tk 100. At any price above Tk 100 she would not buy. At Tk 100 or below she would buy.

Willingness to pay is closely tied to marginal utility, the extra utility from one more unit. The first pot is very useful to the woman. A second pot would add far less utility, so she would pay much less for it. Her willingness to pay therefore falls as she buys more units. This is the reason the demand curve slopes downward: each further unit is worth less to the same consumer.

Two cautions apply. First, willingness to pay is a limit in the mind, not a price on a tag. Nobody pays it unless the price rises to meet it. Second, it differs from person to person. A consumer with

Income is limited and each taka can be spent only once, so every purchase is a choice.

Spend. Word in use. In daily life, to pay out money, or to pass time. In economics, to use part of income to buy goods, so that the same money cannot buy anything else.

Recall: marginal utility, Chapter 3, section 3.1 'The Idea of Utility'.

A sensible consumer spends each taka where it adds the most utility.

Real consumers do not calculate every purchase, so the rule is a guide only.

Recap: consumption, limited income and the gap between price and satisfaction.

The woman's extra Tk 40 needs a name and a measure.

Willingness to pay. The highest price a buyer would pay for one more unit of a good.

Willingness to pay is the highest price a consumer would pay for one more unit.

Recall: marginal utility, Chapter 3, section 3.1; the demand curve, Chapter 4, section 4.1.

Willingness to pay is marginal utility measured in money, and it falls with each extra unit.

Willingness to pay is a limit in the mind, and it differs between consumers.

Two earlier ideas and one appendix help with this chapter.

A diner enjoys her first cup of tea and refuses her fourth.

The tea, the table and the price stayed the same, yet the pleasure changed.

This small puzzle explains why buyers stop buying.

The chapter moves slowly, in small steps, because the ideas are abstract.

Every purchase begins with a want that the buyer hopes to satisfy.

GO SLOWLY HERE.

Utility. The satisfaction a person gets from using a good or service.

Economists call the satisfaction from a good its utility.

Two people can gain different

You will need two ideas from earlier chapters: opportunity cost, from Chapter 2 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost'; and the meaning of an economic law, from Chapter 1 'What Economics Is', section 1.4 'Economic Laws'. Appendix A1 'Maths You Will Use' helps with adding columns of numbers and with reading a graph.

Imagine a long table set for afternoon tea in a Victorian town house. A diner sits down, tired and cold after a walk. The first cup of hot tea is a delight, and she drinks it slowly. The second cup is still pleasant, though the delight is smaller. She drinks the third mainly because it stands in front of her. When the maid offers a fourth, the diner covers the cup with her hand and says, politely, that she has had enough.

Look at what did not change. The tea is the same tea. The cups, the table and the maid are the same. The tea is free, since the hostess pays for it. Yet the drink that gave great pleasure at the start gives almost none at the end. Why should a good thing lose its power to please as we take more of it?

The question looks small, but it is not. Every day, people decide how much of a good to buy and when to stop. A bookseller must ask how many notebooks one customer will want. If every extra notebook pleased as much as the first, a customer would buy without end, and no shop could ever fill the wish. In fact customers stop. To understand why they stop is the first step towards understanding how buyers choose, which later chapters study.

Here we build the answer in small steps. First we name the pleasure and learn to count it, using imaginary numbers. Then we state the rule by which pleasure fades, and we test it on a student buying exercise books at Meghna and Sons. Many students find this chapter slow at first, because satisfaction cannot be seen or weighed. For that reason we go slowly, one idea at a time, with small numbers.

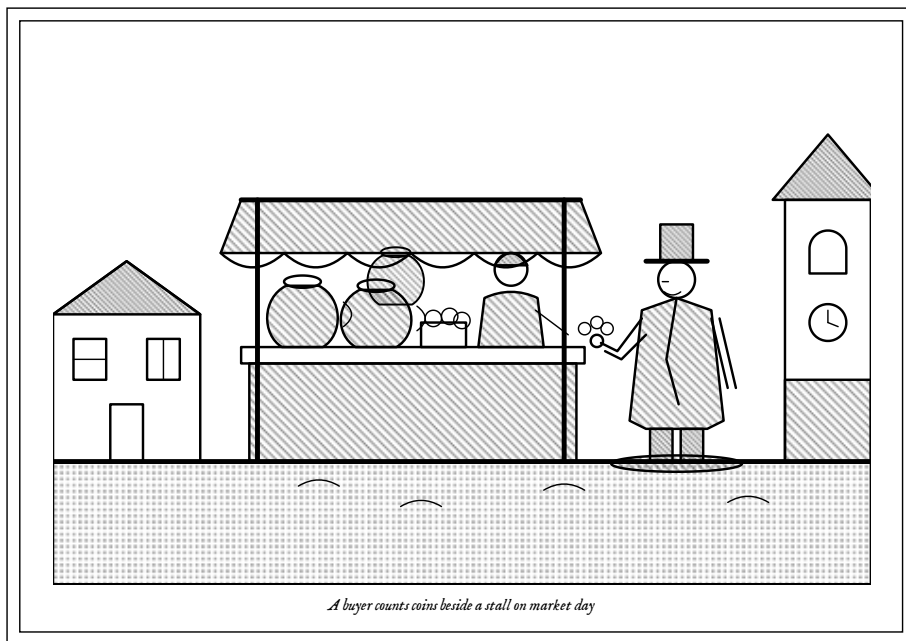
* * *

3.1 THE IDEA OF UTILITY

Begin with your own day. Why did you eat your last meal? You were hungry, and food satisfied the hunger. Hunger is a want, and the pleasure or relief of meeting it is a kind of reward. Every time a person drinks, reads, travels or buys, a want is being met and a reward is felt. Economics needs a short name for that reward, so that we can reason about it without confusion.

Economists call the reward **utility**. Utility means the satisfaction a person gets from using a good or service. A cup of hot tea gives utility to a cold traveller. A bus ticket gives utility to a worker who must reach the office. Notice what the word does not say. Utility is not a judgement of whether a good is wise or healthy. A sweet snack may harm the teeth, yet it gives utility to the person who enjoys it, because utility counts only the satisfaction that person feels.

Utility belongs to the person, not to the good. Picture a market



CHAPTER VI

Consumption and Consumers' Surplus

Why does a consumer who pays the full price still leave with more than she paid for?



This chapter explains how people use goods to satisfy their wants and how they spend a limited income. It then shows that a consumer often gains more from a good than the price paid, and teaches you to measure that gain in taka.

SATISFACTION FROM FIVE CUPS OF TEA (IMAGINARY UTILS)

CUP	MARGINAL UTILITY (UTILS)	TOTAL UTILITY (UTILS)
1	10	10
2	8	18
3	5	23
4	2	25
5	0	25

Read a row to see what one cup added and what all cups gave.

Counting in utils turns a feeling into a list that can be compared.

Utils compare one person's own moments, not different people.

The section ends by recalling the three ideas and the unit.

Three economists in three countries found the same rule within a few years.

Look again at the marginal utility column of the tea table.

GO SLOWLY HERE.

To read a row, take cup 3. Its marginal utility is 5, so the third cup added 5 utils. Its total utility is 23, so the three cups together gave 23 utils. Notice the link between the columns: the total after cup 3 equals the total after cup 2 plus the marginal utility of cup 3, because 18 plus 5 is 23. Each total is the sum of all the marginal utilities so far.

Why count at all? Without numbers, we could say only that the diner liked the first cup more than the second. With numbers, we can say how much more, and we can say how fast the pleasure fades. That speed is exactly what the law in the next section describes. The numbers must be treated as a tool of reasoning, in the way that a map is a tool: it does not equal the land, but it helps a traveller decide.

One limit is important. A util belongs to one person at one time. We may say that the diner liked cup 1 more than cup 2. We may not say that her 10 utils equal 10 utils felt by another person, because no common scale exists.

Before moving on. Utility is satisfaction. Total utility is the satisfaction from all units; marginal utility is the extra from one more unit. We count satisfaction in utils, which are imaginary but useful. You should now be able to say, in your own words, why the total after the fourth cup can be higher than after the third even though the fourth cup added only a little.

3.2 THE LAW OF DIMINISHING MARGINAL UTILITY

In the early 1870s, an economist in England, one in Austria and one in Switzerland each published the same observation about satisfaction. None knew of the work of the others. The section called In History, at the end of this chapter, tells that story. Here we explain the observation itself, and we use the tea table to see it.

THE LAW STATED

Return to the table of five cups. Read down the marginal column: 10, 8, 5, 2, 0. Each number is smaller than the one before. The cups are the same cups, and the diner is the same diner. Yet each extra cup adds less. This pattern is so regular that economists give it a name and call it a law.

The pattern is the **law of diminishing marginal utility**. The

² Recall: laws, Chapter 1, 'What Economics Is', section 1.4 'Economic Laws'. An economic law describes a regular tendency in human choice, not a rule that nature enforces.

THINK IT THROUGH

5.T1 A pharmacy sells a medicine with no substitute, and its price elasticity of demand is 0.3. The owner raises the price by 10%. Estimate the change in quantity demanded and say what happens to total revenue.

5.T2 Explain why a firm whose demand is inelastic might still hesitate to raise its price a great deal. Think of time and substitutes.

WHERE PEOPLE GO WRONG

Many students believe that a steep curve and an inelastic good are the same thing, and that a flat curve always means elastic. Slope depends on the units on the axes; elasticity compares percentages. Along a straight line the slope is constant while elasticity changes from point to point. Students also believe that a negative result means demand is inelastic. The sign shows only that price and quantity move in opposite directions; the size alone tells you which case you have. A third error is to call a good elastic or inelastic everywhere; the result belongs to a price range, a period and a group of consumers. The last error is to use the total revenue test for a change in income. The test applies to the good's own price only.

IN HISTORY

In October 1973 the Yom Kippur War, which began on 6 October between Israel and Egypt and Syria, led Arab members of the Organization of the Petroleum Exporting Countries to announce cuts in production and an embargo on oil shipments to the United States and some other countries. The embargo lasted until March 1974. During this time the price of crude oil rose from about US\$3 a barrel to nearly US\$12, roughly four times.

The rich countries did not respond with a matching fall in the quantity demanded. In the short run, households and firms could not change their cars, heating systems and machines, and there were few substitutes for petrol and heating oil. Demand was therefore strongly inelastic. In the years that followed, consumers bought smaller cars, firms improved their equipment, and new oil fields opened. Demand became more elastic as time passed, as section 5.3 would lead you to expect.

REVIEW QUESTIONS

5.1 Define price elasticity of demand in your own words, and say why percentages are used.

5.2 Why does the midpoint method give the same answer when a price rises as when it falls?

5.3 A price rises from Tk 20 to Tk 30 and the quantity demanded falls from 60 to 40. Calculate E and say whether demand is elastic, inelastic or unit elastic.

Four common misunderstandings, each corrected.

In 1973 oil prices rose fourfold, yet demand fell slowly.

Consumers could not change quickly, so demand fell only slowly.

The case table sets the numbers side by side.

THE STUDENT'S UTILITY FROM EXERCISE BOOKS (UTILS)

BOOK	MARGINAL UTILITY (UTILS)	TOTAL UTILITY (UTILS)
1st	40	40
2nd	30	70
3rd	20	90
4th	10	100
5th	0	100

PAUSE AND CHECK 3.P2 Using the tea table, how much did the fourth cup add to the diner's total satisfaction? Which column gives the answer?

Now we draw the numbers. Every chart in this book has two lines at right angles, called axes.³ On the horizontal axis we place the number of books. On the vertical axis we place utils. A point on the chart means: at this number of books, the utils have this value. The first chart shows total utility.

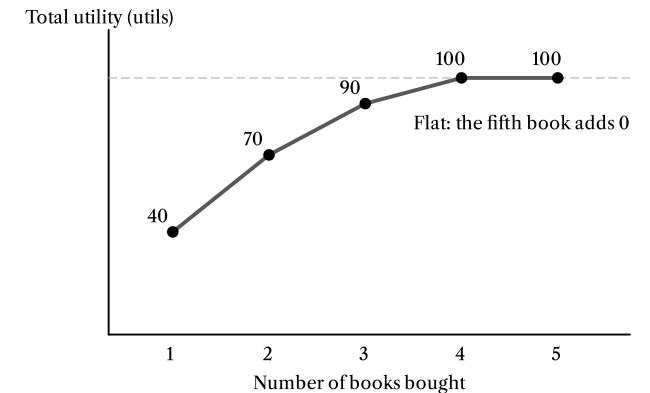


FIG. 3.1. Total utility rises with each book and then becomes flat

Read the chart from left to right. The line climbs steeply from the first book to the second, because 40 utils are added. It climbs less steeply to the third, because 30 and 20 are added. It becomes flat after the fourth book, because the fifth adds nothing. A flat line means the total has stopped rising.

Before reading a chart, learn what its two axes show.

The total line climbs quickly, then more slowly, then stops climbing.

³ See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

Before moving on. A result above 1 is elastic demand, equal to 1 is unit elastic demand, and below 1 is inelastic demand. Total revenue is price times quantity. When demand is elastic a price cut raises it, and when demand is inelastic a price cut lowers it. The same good may be elastic in one price range and inelastic in another.

* * *

5.3 FACTORS BEHIND ELASTICITY

Why is salt inelastic and a pair of shoes elastic? Three features of a good explain most cases. None of them gives a number; each tells you which side of 1 to expect.

The first feature is the availability of substitutes, which are other goods that can be used for the same purpose. A consumer who finds the price of one brand of tea too high can buy another brand, so demand for one brand is elastic. Demand for tea in general has fewer substitutes, so it is less elastic. The more narrowly a good is described, the more substitutes it has.

The second feature is the difference between a necessity, which a household buys whatever the price, and a luxury, which it can easily do without. Salt, rice and basic medicines are necessities. Fashionable shoes and restaurant meals are luxuries. A word of caution: the same good can be a necessity for one household and a luxury for another, so this is a tendency, not a rule.

The third feature has two parts. Time: consumers need time to find substitutes and change habits, so demand is usually less elastic soon after a price change than long after. Share of income: a good that takes a large part of a household's income, such as rent, makes consumers watch its price closely, while a pen takes so little that its price rarely matters. Both parts shift the result toward elastic as time passes or the share rises.

Before moving on. Demand tends to be more elastic when close substitutes exist, when the good is a luxury, when consumers have time to adjust, and when the good takes a large share of income. Each is a guide to which side of 1 to expect.

* * *

5.4 INCOME AND CROSS ELASTICITY

The method is not limited to the good's own price. Quantity demanded also responds to consumers' income and to the price of other goods. Each response has its own measure, found in the same way.

Income elasticity means the percentage change in quantity demanded divided by the percentage change in consumers' income. A change in income shifts the demand curve rather than moving along it, which is why this measure is separate from the price measure. The sign matters here and is kept. A positive result means that more income raises the quantity demanded; a negative result means that it lowers it.

Before moving on: the result is read against 1, and the revenue test follows from it.

Three questions help predict whether demand is elastic or inelastic.

Many close substitutes make demand elastic.

Necessities tend to be inelastic and luxuries elastic.

Time and the share of income also matter.

Before moving on: substitutes, necessity and luxury, time and share of income.

The same idea can measure responses to causes other than the good's own price.

Recall: movement and shift, Chapter 4, section 4.2 'Movement and Shift'.

Income elasticity. The percentage change in quantity demanded divided by the percentage change in consumers' income.

MU(n) is the marginal utility of the nth unit. Rearranged, $MU(n) = TU(n) - TU(n-1)$.⁴

Example 1. *Given:* the marginal utilities of the first three exercise books are 40, 30 and 20 utils. *Find:* the total utility after three books. *Formula:* total utility is the sum of the marginal utilities so far. *Steps:* $40 + 30 = 70$ utils; $70 + 20 = 90$ utils. *Answer:* total utility after three books is 90 utils. *Check:* the case table shows 90 in the row of the 3rd book, so the answer is sensible.

Example 2. *Given:* a different person drinks glasses of sugarcane juice, and total utility after each glass is 12, 22, 30, 36, 40, 42 and 42 utils. *Find:* the marginal utility of each glass. *Formula:* $MU(n) = TU(n) - TU(n-1)$, and for the first glass $MU(1) = TU(1)$. *Steps:* glass 1: 12 ; glass 2: $22 - 12 = 10$; glass 3: $30 - 22 = 8$; glass 4: $36 - 30 = 6$; glass 5: $40 - 36 = 4$; glass 6: $42 - 40 = 2$; glass 7: $42 - 42 = 0$. *Answer:* the marginal utilities are 12, 10, 8, 6, 4, 2 and 0 utils. *Check:* the list falls at every step, as the law predicts, and the total stops rising when marginal utility reaches zero.

Common slips. Students often read the marginal column as if it were the total column. They forget to write utils beside the numbers. They subtract in the wrong order, which gives a negative number for a gain. And they treat the first row differently from the others: for the first unit, marginal utility and total utility are the same number.

SUMMARY

Utility is the satisfaction a good gives. Total utility is the whole satisfaction from all units, and marginal utility is the extra from one more unit. We count satisfaction in imaginary utils. The law of diminishing marginal utility says that each extra unit adds less than the one before. Total utility rises as long as marginal utility is above zero, and it stops rising when marginal utility reaches zero.

THINK IT THROUGH

3.T1 Why might the diner's fifth cup of tea give her zero utility? Give two reasons from the diner's own situation.

3.T2 In the tea table, suppose the diner could drink a sixth cup that gave her minus 3 utils. What would her total utility be after six cups, and what should she do?

WHERE PEOPLE GO WRONG

Error 1: "If marginal utility falls, the total falls." This is wrong. A smaller extra is still an extra, so the total rises while marginal utility is above zero. The total falls only when marginal utility is negative.

Error 2: "Utility means a good is useful or good for you." This is wrong. Utility is the satisfaction a person feels, whether or not the good is wise. Economics describes the feeling and leaves the moral judgement to others.

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

The first example uses the student's own numbers and finds a total.

The second example starts from totals and finds the extra from each unit.

Four common slips spoil these calculations.

The chapter's main ideas in a few lines.

Two short problems that use the tea table.

A common error is to believe that falling marginal utility means falling total utility.

Utility is not the same as usefulness or goodness.

5.2 READING THE RESULT

A number such as 1.29 is useful only when you know what it says. The dividing line is 1, because 1 is the case where the two percentages are equal. Three cases follow, each with a name.

When the result is above 1, demand is **elastic demand**. It means demand for which the percentage change in quantity is larger than the percentage change in price. The notebook, at 1.29, is an example. The tea stall at 2 is another. The curve for such a good is flat, as the first figure below shows.

When the result is below 1, demand is **inelastic demand**. It means demand for which the percentage change in quantity is smaller than the percentage change in price. Salt in our opening puzzle is a likely example: a price cut of one-fifth brings only a trifling rise in the quantity demanded. The curve for such a good is steep.

When the result equals 1, demand is **unit elastic demand**. It means demand for which the percentage change in quantity equals the percentage change in price. Your pause question, with a result of 1, is an example. It is the dividing line between the other two cases.

To read the figure below, notice what the axes show. The vertical axis shows price, and the horizontal axis shows quantity demanded. Each point on a curve is one price with the quantity demanded at that price. Starting from the same point, a price fall of the same size moves the flat curve a long way to the right and the steep curve only a short way.

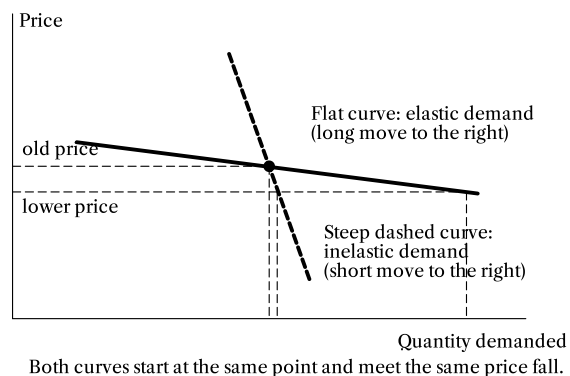


FIG. 5.1. Two demand curves through the same point: a flat curve for elastic demand and a steep curve for inelastic demand

A warning follows. Elasticity is not the same as the slope of the curve. Along one straight demand curve, the slope never changes, yet the elasticity does. On the notebook schedule, the fall from Tk

A single number is useful only if you know what it says.

Elastic demand. Demand for which the percentage change in quantity is larger than the percentage change in price.

Elastic demand has a result above 1: consumers respond strongly.

Inelastic demand. Demand for which the percentage change in quantity is smaller than the percentage change in price.

Inelastic demand has a result below 1: consumers respond weakly.

Unit elastic demand. Demand for which the percentage change in quantity equals the percentage change in price.

Unit elastic demand has a result of exactly 1: the two percentages are equal.

The axes show price and quantity demanded; each point is one pair.

Slope and elasticity are different things, as a straight line shows.

This measure has a name. **Price elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in price. Suppose a price rises by 10% and the quantity demanded falls by 20%. The price elasticity of demand is 20 divided by 10, which is 2. The name comes from a picture: a rubber band is elastic when a small pull stretches it far, and quantity demanded is elastic when a small change in price stretches it far.¹

Before moving on. Changes in money and in units cannot be compared, so we use percentages. Price elasticity of demand divides the percentage change in quantity demanded by the percentage change in price. A result of 2 means quantity changes twice as fast as price.

FINDING A PERCENTAGE CHANGE

You need one tool before you can calculate. In words, the percentage change equals the change in the value, divided by the value you compare it with, multiplied by 100. In symbols, percentage change = (new value - old value) ÷ old value × 100. This ordinary method works well for small changes. It has one weakness, which many students find confusing, because the answer depends on the direction of travel.

Test it with the price of a notebook. A price rising from Tk 40 to Tk 50 rises by Tk 10, and 10 divided by 40 is 25%. A price falling from Tk 50 to Tk 40 falls by the same Tk 10, but 10 divided by 50 is 20%. One gap between the same two prices has produced two percentages. If we used this method, the elasticity between Tk 40 and Tk 50 would depend on which price we chose as the start, and two people measuring the same demand schedule could report different numbers.

The remedy is the midpoint method, which divides the change by the midpoint, the average of the old and new values. In words: percentage change = change ÷ average of the old and new values × 100. In symbols, with old value A and new value B, percentage change = (B - A) ÷ ((A + B) ÷ 2) × 100. The average is the same whichever value you call old, so the answer is the same in both directions. Between Tk 40 and Tk 50 the average is Tk 45, the change is Tk 10, and 10 ÷ 45 gives 22.22% whether the price rises or falls.²

Before moving on. The ordinary percentage change depends on the starting point. The midpoint method removes that problem by

¹ By the law of demand, price and quantity demanded move in opposite directions, so the raw ratio is negative: -20 divided by +10 gives -2. Economists usually drop the minus sign and report the size of the number, ignoring its sign. This book does the same, and the sign is always understood to be there.

² Some books divide by the starting value and accept a small difference. The midpoint method is used here because it gives one answer for each pair of points. If another book uses the starting value, expect slightly different results when the changes are large.

Elastic. Word in use. In daily life, able to stretch and then return, like a rubber band. In economics, the quantity demanded stretches a long way when the price changes.

Price elasticity of demand.

The percentage change in quantity demanded divided by the percentage change in price.

Price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price.

Before moving on: the measure compares two percentages and has no units.

A percentage change is the change divided by a base value, times 100.

The ordinary method gives different answers going up and going down.

The midpoint method divides each change by the average of the old and new values.

Before moving on: use the average of old and new values as the base.

Two earlier ideas and one appendix help with this chapter.

A fishmonger changes a price tablet, and the crowd responds.

The crowd does not obey the fishmonger's orders, yet it behaves in a regular way.

The chapter gives that pattern a name, a table and a picture.

Begin with a question: how many notebooks would you buy at each price?

GO SLOWLY HERE.

Demand. The quantities of a good that buyers are willing and able to buy at each price in a period.

Demand is the whole set of prices and quantities, not one number.

Demand. Word in use. In daily life, demand means a strong request or order, as in a demand for payment. In

You will need two ideas from earlier chapters: the fall of satisfaction with each extra unit, from Chapter 3 'Utility and Diminishing Marginal Utility', section 3.2 'The Law of Diminishing Marginal Utility'; and the habit of holding other conditions fixed, from Chapter 1 'What Economics Is', section 1.4 'Economic Laws'. Appendix A1 'Maths You Will Use' helps with reading a graph and with rearranging an equation.

Picture a fishmonger's stall in a Victorian market town, early on a Saturday morning. Along the front of the stall stand small tablets, each marking the price of one kind of fish. A crowd of buyers moves slowly past. At nine o'clock the fishmonger lowers the tablet for the herring. Within minutes the crowd thickens, baskets fill, and the tray empties faster than before. At noon the tablet is raised again, and the crowd moves on to the cheaper trays.

Nobody ordered the buyers to do this. Each acted for private reasons, and each had a different purse and a different taste. Yet the crowd as a whole behaved in a way that the fishmonger could predict: a lower price brought more buyers, and a higher price sent them away. How can millions of separate choices form a pattern that is so regular? And can we describe the pattern exactly, so that any firm could use it?

In this chapter we build the answer in small steps. We name what buyers do, record it in a table, and draw it as a curve. Then we ask what makes the curve move, and we meet some goods that behave in unusual ways. Many students find the central diagram hard at first, because two different changes both seem to alter the same picture. We therefore go slowly, and we separate the two changes with care.

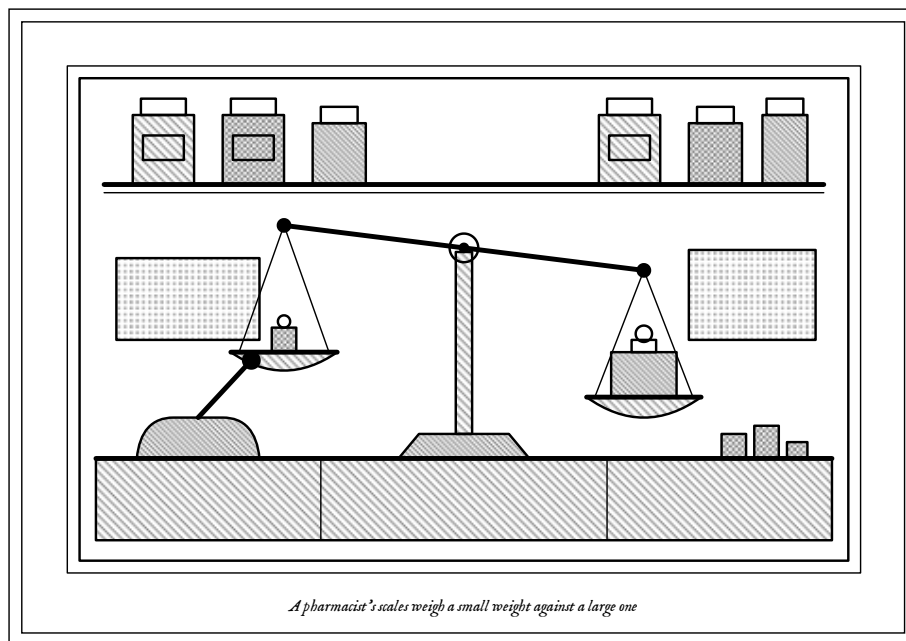
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4.1 QUANTITY DEMANDED AND THE LAW OF DEMAND

Begin with a question from your own life. A plain 200-page notebook costs Tk 60. How many would you buy this week? Now suppose the price falls to Tk 20. Would you buy the same number? Most people answer that they would buy more at the lower price, and fewer at the higher price. The number of units that buyers choose to buy at one particular price in a period is called the quantity demanded. It is a single number, tied to a single price.

Economists use the word **demand** for something larger. Demand means the quantities of a good that buyers are willing and able to buy at each price in a period. Notice the two conditions, willing and able. A wish without money is not demand, and money without a wish is not demand either. For the notebook, demand is the full list: so many at Tk 60, so many at Tk 50, so many at Tk 40, and so on. One entry in the list is a quantity demanded. The whole list is demand.

The everyday word needs care here. In daily speech people say that "demand rose" when they mean that more was bought. This book keeps two words apart. Quantity demanded means one



CHAPTER V

Elasticity of Demand

How hard does a change in price push the consumer?



This chapter teaches you to measure how strongly consumers respond to a change in price, in their own income and in the price of a rival good, and to use the result to judge whether a lower price brings a firm more money.

The line slopes downward, because a lower price goes with a larger quantity.

The section closes with the four ideas in a few lines.

Two different changes can alter the picture, and they must be kept apart.

GO SLOWLY HERE.

A change in the price of the good moves us along one curve.

Five other conditions decide where the whole curve lies.

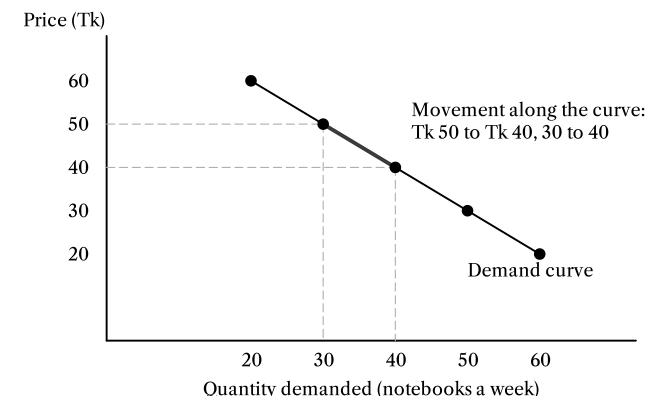


FIG. 4.1. The demand curve for the notebook, with a movement along it from Tk 50 to Tk 40

Read the line from the top left to the bottom right. At the top, the price is high and the quantity is small. At the bottom, the price is low and the quantity is large. We call it a curve although in this case it is a straight line, because the word covers lines of every shape. The downward slope is the law of demand drawn as a picture: lower price, larger quantity demanded. In the picture, the short bold part between the points for Tk 50 and Tk 40 will be used in the next section.

Before moving on. Quantity demanded is one number at one price. Demand is the whole list of prices and quantities, which the demand schedule records and the demand curve draws. The law of demand says that, other conditions staying the same, a lower price brings a larger quantity demanded. You should now be able to say why the curve slopes downward.

4.2 MOVEMENT AND SHIFT

The fishmonger faced two different events. At nine o'clock his own price fell, and more fish sold. Suppose that on another Saturday a rival stall across the street cut its price, and fewer buyers came to him even though his tablet had not changed. Both events changed how much fish he sold. Are they the same kind of change? They are not, and the difference is the key to this chapter.

First, the price of the good itself changes. The notebook falls from Tk 50 to Tk 40. Buyers respond by buying more, and the quantity demanded rises from 30 to 40 notebooks a week. On the graph, the point moves along the same line, from the point for Tk 50 to the point for Tk 40. We call this a movement along the curve. The curve itself does not change position, because the whole list of prices and quantities is the same list as before.

Second, something other than the price changes. The curve was drawn with other conditions held fixed, and these conditions are called the determinants of demand. When one of them changes, buyers want different quantities at every price, and the curve must

is one number at one price. Demand alone is the whole list of prices and quantities.

Error 3: "Sales rose after the price rose, so the law of demand fails." This may be wrong. Incomes, tastes or the price of a rival may also have changed, and these shift the curve. The law speaks only of the effect of the good's own price when other conditions stay the same.

Error 4: "A buyer who wishes for a good demands it." This is wrong. Demand needs both the will and the ability to pay.

IN HISTORY

In 1838, the French mathematician Antoine Augustin Cournot, who lived from 1801 to 1877, published *Recherches sur les principes mathématiques de la théorie des richesses*, which is known in English as *Researches into the Mathematical Principles of the Theory of Wealth*. In its fourth chapter, titled in English "Of the law of demand", Cournot wrote demand as a function of price: the quantity sold depends on the price. He treated it as a fact of experience that the function falls as the price rises, and he drew it as a curve with price and quantity as its two axes.

Cournot's book received hardly any response when it appeared, and the reading public of the 1830s had little interest in economics written with algebra and diagrams. Later economists, among them Léon Walras of Lausanne, drew on his ideas. Today every textbook, including this one, draws the demand curve in the way that Cournot first did.

REVIEW QUESTIONS

- 4.1 Plot the demand schedule of the notebook from the case file on a graph, with price on the vertical axis. Say what the slope of the line shows.
- 4.2 After the rival's price fall, the quantities demanded at Tk 60, 50, 40, 30 and 20 are 15, 25, 35, 45 and 55. Plot the new curve on the same graph as the old one. Then draw a third curve that shows a rise in demand of 10 at every price, and give its quantity at Tk 40.
- 4.3 Classify each change as a movement along the curve or a shift of the curve: (a) the notebook price falls from Tk 60 to Tk 50; (b) pens become cheaper; (c) incomes of students fall; (d) the shop lowers the notebook price from Tk 40 to Tk 30.
- 4.4 Using $Q = 80 - P$, find the quantity demanded at Tk 35 and the price at which the quantity demanded is 55.
- 4.5 Name a pair of substitute goods and a pair of complementary goods. For each pair, say how a price rise for the first good changes demand for the second.
- 4.6 Define a Giffen good, and say why it is not the usual case.
- 4.7 A new college opens near the shop, and the price of the notebook is unchanged. Say what happens to demand, and explain which determinant has changed.

The law needs other things to stay equal.

Wanting is not the same as demanding.

A French mathematician drew the first demand curve.

The work was ignored for years, but its method became standard.

Seven questions that check the chapter's ideas.

4.3 RELATED AND UNUSUAL GOODS

Some changes in demand come from the price of another good.

Substitute goods. Goods that can replace each other, so a price rise for one raises demand for the other.

Substitute goods can replace each other.

Complementary goods. Goods used together, so a price rise for one lowers demand for the other.

Complementary goods are used together.

Read each row as a price change for the first good and its effect on the second.

Normal good. A good for which demand rises when buyers' incomes rise.

A normal good is bought more when incomes rise.

Normal. Word in use. In daily life, normal means usual or ordinary. In economics, a normal good is one whose demand rises with income.

The economic word normal has a narrow meaning.

Giffen good. A rare, very basic good for which quantity demanded rises when its price rises.

A Giffen good breaks the law

A shift can come from a change in the price of another good, and the effect depends on how the two goods are related. The rival notebook in the case file is the first kind of relation. A shopper who finds the rival cheaper may switch to it, and our notebook loses buyers. The second kind of relation works the other way. We look at both kinds now, then at income, and then at four unusual cases.

Two notebooks of the same size and quality are **substitute goods**. Substitute goods are goods that can replace each other, so a price rise for one raises demand for the other. If the rival notebook becomes dearer, some buyers move to ours, and demand for ours shifts to the right. The case file showed the reverse: a price fall for the rival shifted our demand to the left. Tea and coffee, or buses and trains, are other pairs.

Some goods are not rivals but partners. **Complementary goods** are goods used together, so a price rise for one lowers demand for the other. A pen and a notebook are complementary goods. If pens become dearer, fewer people write, and fewer notebooks are wanted at every price. Tea and sugar, and a lamp and its oil, are other pairs. Substitutes move in opposite directions, and complements move in the same direction, as the table below summarises in words first.

HOW A PRICE RISE FOR ONE GOOD CHANGES DEMAND FOR ANOTHER

RELATION	EXAMPLE PAIR	PRICE OF FIRST GOOD	DEMAND FOR SECOND GOOD
Substitute goods	Our notebook and a rival notebook	Rises	Rises
Complementary goods	Pen and notebook	Rises	Falls

To read the second row, take the pair of pen and notebook. When the price of the pen rises, demand for the notebook falls. Be careful with direction: the effect on the second good is a shift of its curve, not a movement along it.

Income is another determinant. A **normal good** is a good for which demand rises when buyers' incomes rise. Notebooks are a normal good, because families with more money buy more of them. Most goods behave in this way. A few goods behave in the opposite way, and demand for them falls as incomes rise, because buyers move to better goods; we do not name them here.

The word needs care. A normal good is not a usual good, nor an average one. It is any good whose demand rises when buyers' incomes rise. A luxury can be a normal good, and so can a basic good.

The law of demand has a famous exception. A **Giffen good** is a rare, very basic good for which quantity demanded rises when its price rises. Alfred Marshall, an English economist, wrote in 1890 that Sir Robert Giffen, a Victorian statistician, had noticed such

more along the supply curve. The market settles at a higher price and a larger quantity.

Case file, illustrative. In the notebook market, demand rises to a quantity demanded of 100 minus the price, and supply stays at a quantity supplied equal to the price. At the new equilibrium, $100 - \text{price} = \text{price}$, so $100 = 2 \times \text{price}$ and the price is Tk 50. The quantity is 50 notebooks. The price has risen from Tk 40 to Tk 50, and the quantity from 40 to 50.

PAUSE AND CHECK 9.P2 If demand shifts to the right and supply does not shift, what happens to the equilibrium price and the equilibrium quantity?

Now take a supply shift. When the price of paper rises, firms make notebooks only at a higher cost to themselves, and the supply curve shifts to the left. At the old price, buyers want more than firms offer, so the price rises. As it rises, buyers move up the demand curve and buy fewer notebooks. The new equilibrium has a higher price and a smaller quantity.

Case file, illustrative. In the notebook market, supply falls to a quantity supplied of the price minus 10, and demand returns to the original quantity demanded of 80 minus the price. At the new equilibrium, $80 - \text{price} = \text{price} - 10$, so $90 = 2 \times \text{price}$ and the price is Tk 45. The quantity is 35 notebooks. The price has risen from Tk 40 to Tk 45, and the quantity has fallen from 40 to 35.

The table below collects the four single shifts. Read each row from the left: the first column names the shift, and the next two columns give the effect on the equilibrium price and the equilibrium quantity.

EFFECTS OF SINGLE SHIFTS ON EQUILIBRIUM

SHIFT	EQUILIBRIUM PRICE	EQUILIBRIUM QUANTITY
Demand shifts right	Rises	Rises
Demand shifts left	Falls	Falls
Supply shifts right	Falls	Rises
Supply shifts left	Rises	Falls

Sometimes both curves shift at once. Then the table gives two answers that may disagree. If demand shifts right and supply shifts left, both shifts push the price up, so the price certainly rises. But the shifts push the quantity in opposite directions, so the quantity depends on which shift is larger. When the effects disagree, we must calculate with the numbers; guessing is unsafe.

Before moving on. A shift moves a whole curve; a price change moves us along a curve. Demand shifting right raises both price and quantity. Supply shifting left raises the price and lowers the quantity. With two shifts, one effect is certain and the other depends on the sizes.

* * *

WORKED EXAMPLES

The case file finds the new crossing after demand rises.

When supply shifts left, sellers offer less at every price, so price rises and quantity falls.

The case file finds the new crossing after supply falls.

Four single shifts have four predictable results, shown in one table.

When both curves shift together, one effect is certain and the other depends on size.

Before moving on, recall the shift rule and the two kinds of change.

Consumers' surplus. The gap between what buyers are willing to pay for a good and what they actually pay.

Surplus. Word in use. In daily life, an amount left over after needs are met. In economics, the gain a consumer keeps when the price paid is below the most the consumer would pay.

Consumers' surplus is the gap between willingness to pay and the price paid.

Three of five customers buy the notebook at Tk 40, and their surplus totals Tk 30.

The customer sees a bargain, but the owner sees only a Tk 40 sale.

To measure surplus, subtract the price from willingness to pay for each unit bought, then sum.

Read the table row by row: the first buyer gains most and the last buyer gains least.

The surplus depends on the price: a lower price gives the same consumer a larger surplus.

Now the puzzle has an answer. **Consumers' surplus** means the gap between what buyers are willing to pay for a good and what they actually pay.¹ The woman's willingness to pay was Tk 100 and she paid Tk 60, so her consumers' surplus on the pot is Tk 100 minus Tk 60, which is Tk 40. The surplus is not money. It is satisfaction measured in taka, and it never appears on a receipt.

Case file, illustrative. Five regular customers of Meghna and Sons were recorded with these willingness to pay amounts for the popular 200-page notebook: Tk 60, Tk 50, Tk 40, Tk 30 and Tk 20. The market price is Tk 40. Three customers buy, because the price is at or below their willingness to pay. Consumers' surplus = $(60 - 40) + (50 - 40) + (40 - 40) = \text{Tk } 30$.

The same sale looks different from each side. The customer with a willingness to pay of Tk 60 sees a bargain of Tk 20. The owner sees a sale at Tk 40, the same as every other sale of the notebook that day. The owner cannot see the Tk 20, because it exists only in the mind of the customer.

The method has three steps. For each unit bought, subtract the price from the willingness to pay. Omit any unit whose willingness to pay is below the price, because the consumer does not buy it. Then sum the differences. The table applies the steps to the five customers of the notebook case. Its columns show the customer, the willingness to pay, whether the customer buys at Tk 40, and the surplus.

NOTEBOOK CASE AT THE PRICE OF TK 40

CUSTOMER	WILLINGNESS TO PAY (TK)	BUYS AT TK 40?	SURPLUS (TK)
A	60	Yes	20
B	50	Yes	10
C	40	Yes	0
D	30	No	0
E	20	No	0
Total			30

Read the row for Customer A first. A would pay Tk 60 and pays Tk 40, so A keeps Tk 20. Customer C buys but keeps nothing, because the willingness to pay equals the price. Customers D and E do not buy and gain no surplus. The first buyer has the largest surplus and the last buyer the smallest, which follows from a willingness to pay that falls with each unit.

One more point needs care. The surplus belongs to the consumer and the price together, not to the good alone. The same customer who keeps Tk 20 at a price of Tk 40 would keep Tk 30 at a price of Tk 30. A consumer whose willingness to pay equals the ¹The idea is older than its name. The section 'In History', at the end of this chapter, tells how it began.

shows one price and the quantity that goes with it. The demand curve slopes downward and the supply curve slopes upward. Where they cross, the quantity demanded equals the quantity supplied, and that crossing is the equilibrium.

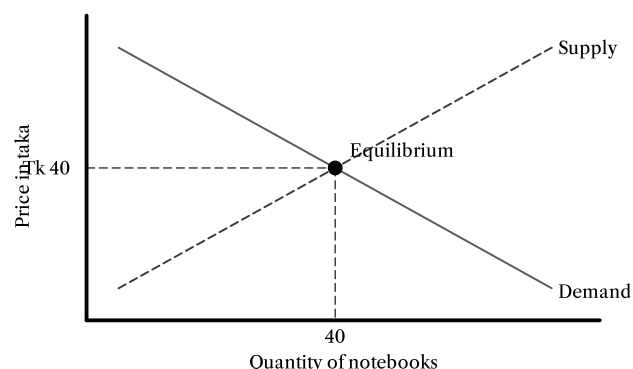


FIG. 9.1. Notebook market: demand and supply cross at the equilibrium price

PAUSE AND CHECK 9.P1 At the equilibrium price of Tk 40, how many notebooks do buyers want, and how many do firms offer?

Now ask what happens at a price other than Tk 40. The table below lists three prices from the notebook market. For each row, compare the two middle columns, which show how many notebooks buyers want and how many firms offer.

NOTEBOOK MARKET AT THREE PRICES

PRICE	QUANTITY DEMANDED	QUANTITY SUPPLIED	RESULT
Tk 30	50	30	Buyers want 20 more than firms offer
Tk 40	40	40	The two quantities are equal
Tk 50	30	50	Firms offer 20 more than buyers want

At Tk 30, buyers want 50 notebooks but firms offer only 30. This is **excess demand**. Excess demand means a situation in which buyers want more than sellers offer at a price, pushing the price up. The 20 buyers who find empty shelves offer more money to win the notebooks, and sellers notice that they can charge more. The price therefore rises towards Tk 40.

At Tk 50, firms offer 50 notebooks but buyers want only 30. This is **excess supply**. Excess supply means a situation in which sellers offer more than buyers want at a price, pushing the price down. Twenty notebooks stay unsold, and sellers cut the price to empty their shelves. The price therefore falls towards Tk 40.

GO SLOWLY HERE.

What happens at any other price is the work of the next two ideas.

Excess demand. A situation in which buyers want more than sellers offer at a price, pushing the price up.

Too low a price leaves buyers wanting more than is offered.

Excess supply. A situation in which sellers offer more than buyers want at a price, pushing the price down.

Too high a price leaves sellers with unsold goods.

A fall in price is a movement along the demand curve, not a shift, and it raises surplus.

The notebook price falls to Tk 30: four customers buy and surplus rises to Tk 60.

Surplus rises for two reasons: old buyers gain more per unit, and new buyers join.

A firm cannot collect the surplus without raising the price, and a higher price reduces the quantity bought.

Recap: uses, limits and the effect of a fall in price on surplus.

Two examples apply the method: the first with small numbers, the second with more realistic ones.

Example 1: surplus on three cups of tea at Tk 6 a cup.

The formula subtracts the price from each willingness to pay and sums the results.

The surplus is Tk 6, and the check agrees.

because the first units give a very large surplus that no chart can measure sensibly.

Now consider a change in price. When the price of the notebook falls, consumers move along the demand curve to a larger quantity demanded. This is a movement along the curve. It is not a shift, which needs a change in something other than price, such as income. Our question is what the movement does to the surplus.

Case file, illustrative. The market price of the notebook falls from Tk 40 to Tk 30. Four of the five customers now buy. Consumers' surplus = $(60 - 30) + (50 - 30) + (40 - 30) + (30 - 30) = 30 + 20 + 10 + 0 = \text{Tk } 60$.

The surplus rose from Tk 30 to Tk 60, a rise of Tk 30. Two things explain it. First, each of the three old buyers keeps Tk 10 more, because the price fell by Tk 10, and $3 \times \text{Tk } 10$ is Tk 30. Second, a new buyer joins. In this case the new buyer adds nothing, because the willingness to pay of Tk 30 equals the new price. With thousands of customers, new buyers join in a steady stream, and each adds a thin slice to the triangle.

A manager must take care here. The surplus is a gain of the consumers, not income of the firm. The firm could collect more of it only by raising the price, and a higher price lowers the quantity demanded. Some consumers would then stop buying. So the surplus is a fact to understand, and it does not tell the manager what price to charge.

Before moving on. Surplus shows the value of a good beyond its price, and governments and firms can use it. It is measured in money, so it cannot compare the feelings of different people exactly, and it can only be estimated. A fall in price is a movement along the demand curve, and it raises the consumers' surplus.

* * *

WORKED EXAMPLES: MEASURING CONSUMERS' SURPLUS

We now apply the method in full, in the order Given, Find, Formula, Steps, Answer and Check. The first example uses small numbers, so that the method is the only new thing to learn. The second is a little messier and shows a change in price.

Example 1: cups of tea. *Given.* A consumer is willing to pay Tk 10 for the first cup of tea, Tk 8 for the second, Tk 6 for the third and Tk 4 for a fourth. The price is Tk 6 a cup. *Find.* The consumers' surplus.

Formula. In words: consumers' surplus equals the sum, over the units bought, of the willingness to pay minus the price. In symbols: $S = (W_1 - P) + (W_2 - P) + \dots$, where S is the consumers' surplus, W is the willingness to pay for each unit bought and P is the price. **Steps.** The fourth cup is not bought, because Tk 4 is below the price of Tk 6. First cup: $10 - 6 = \text{Tk } 4$. Second cup: $8 - 6 = \text{Tk } 2$. Third cup: $6 - 6 = \text{Tk } 0$. Sum: $4 + 2 + 0 = \text{Tk } 6$.

Answer. The consumers' surplus is Tk 6. **Check.** The consumer pays $3 \times \text{Tk } 6 = \text{Tk } 18$ and would have paid up to $10 + 8 + 6 = \text{Tk } 24$.

MEASURING IT

Measuring needs one decision: a percentage of which number? If the price moves from Tk 30 to Tk 40, the rise is 33% of Tk 30 but only 25% of Tk 40. The same journey gives two answers. The *midpoint method* removes the problem by dividing each change by the average of the start value and the end value.

The formula, in words: elasticity of supply equals the change in quantity supplied divided by the average quantity, all divided by the change in price divided by the average price. In symbols: $E_s = (\Delta Q \div \text{average } Q) \div (\Delta P \div \text{average } P)$. Here E_s is the elasticity of supply, ΔQ is the change in quantity supplied, and ΔP is the change in price.¹

Case file, illustrative. In the notebook market where Meghna and Sons sells its popular 200-page notebook, the quantity supplied equals the price in taka. At a price of Tk 30, firms offer 30 notebooks. At a price of Tk 40, firms offer 40 notebooks. We now calculate the elasticity of supply between these two prices.

Given. The price rises from Tk 30 to Tk 40. The quantity supplied rises from 30 to 40 notebooks. *Find.* The elasticity of supply between Tk 30 and Tk 40. *Formula.* $E_s = (\Delta Q \div \text{average } Q) \div (\Delta P \div \text{average } P)$, as defined above. *Steps.* (1) Change in quantity: $40 - 30 = 10$ notebooks. (2) Average quantity: $(30 + 40) \div 2 = 35$ notebooks. (3) Change in quantity as a fraction of the average: $10 \div 35 = 0.286$. (4) Change in price: $40 - 30 = \text{Tk } 10$. (5) Average price: $(30 + 40) \div 2 = \text{Tk } 35$. (6) Change in price as a fraction of the average: $10 \div 35 = 0.286$. (7) Divide: $0.286 \div 0.286 = 1.0$. *Answer.* The elasticity of supply between Tk 30 and Tk 40 is 1.0. *Check.* Both fractions are equal, so their ratio must be 1.

TYPES OF ELASTICITY OF SUPPLY

The size of the number places supply into one of five types. Read the table row by row: the first column gives the value, the second names the type, and the third says what firms do when the price rises.

FIVE TYPES OF ELASTICITY OF SUPPLY

VALUE	TYPE	WHAT FIRMS DO WHEN THE PRICE RISES
Greater than 1	Elastic supply	Quantity supplied rises by a larger percentage than the price
Equal to 1	Unit elastic supply	Quantity supplied rises by the same percentage as the price
Between 0 and 1	Inelastic supply	Quantity supplied rises by a smaller percentage than the price
Zero	Perfectly inelastic supply	Quantity supplied does not change; the supply curve is a vertical line
Without limit	Perfectly elastic supply	Firms offer any quantity at one price; the supply curve is a horizontal line

¹ Some books divide by the starting value instead of the average. That method gives a slightly different number when the change is large. This book uses the midpoint throughout.

Percentage of which number? The midpoint answers this question.

The formula, first in words and then in symbols.

The case file gives a clean pair of numbers for the first calculation.

Each step of the first calculation is shown, in the standard order.

The size of the number places supply into one of five types.

A fourth mistake is to call a fall in price a shift.

A fifth mistake is to confuse the triangle with the rectangle on the chart.

In 1844 Jules Dupuit, a French engineer, showed that people gain more from a bridge than the toll they pay.

Dupuit saw that a lower toll brings more crossings and a larger gain; Marshall later named the idea.

Governments still use the idea to compare a project's benefits with its cost to build.

A fourth mistake is to call a fall in price a shift. A fall in the price of a good moves consumers along the same demand curve to a larger quantity demanded. A shift needs a change in something else, such as income.

A fifth mistake is to confuse the two parts of the chart. The rectangle shows the money that consumers pay, and the firm receives it. The triangle shows the surplus, and the consumers keep it. A reader who adds the two together has found the total value of the units bought to consumers, which is a different quantity from the surplus.

IN HISTORY

In 1844 a French engineer named Jules Dupuit published an article on the utility of public works in a French engineering journal, the *Annales des ponts et chaussées*.² He worked on roads and bridges for the state, and he asked how the benefit of such works should be judged. He showed that people gain more from a bridge than the toll they pay.

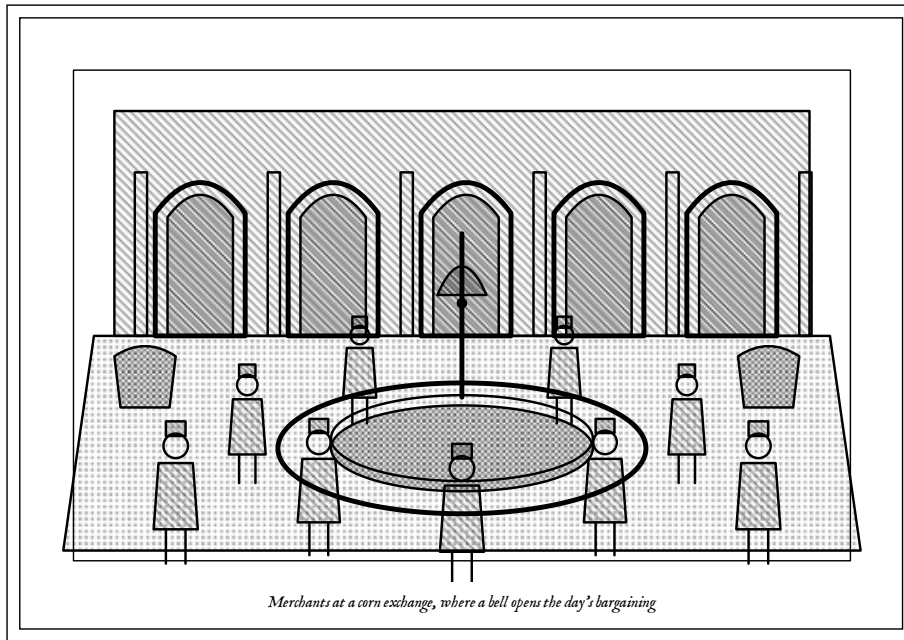
Dupuit saw that a lower toll brings more crossings, and that those who cross at a high toll would have paid even more. He called the gain *relative utility*. The British economist Alfred Marshall, who taught at Cambridge, later gave the idea the name used today. He did so in his textbook *Principles of Economics*, first published in 1890.

Governments still use the idea. When they compare a bridge, a road or a railway with the cost to build it, they include the surplus of the people who will use it. The cost here is the builder's expense, and the benefit is more than the tolls collected.

REVIEW QUESTIONS

- 6.1 Define consumption in your own words and give one example from your own day.
- 6.2 Explain the difference between price and willingness to pay, using the notebook case.
- 6.3 In the notebook case, compute the consumers' surplus when the price is Tk 50, and again when the price is Tk 20.
- 6.4 Four consumers are willing to pay Tk 30, Tk 22, Tk 16 and Tk 10 for one pen each. Compute the consumers' surplus at a price of Tk 16 and at a price of Tk 10, and find the rise between them.
- 6.5 Using marginal utility and the demand curve, explain why the surplus on the last unit a consumer buys is close to zero.

² Dupuit, 'De la mesure de l'utilité des travaux publics', *Annales des ponts et chaussées*, 1844. An English translation appeared in *International Economic Papers* in 1952.



CHAPTER IX

Elasticity of Supply and Market Equilibrium

Who decides a price when no single person decides it?



This chapter measures how strongly sellers respond when the price changes. It then joins the buyers' side and the sellers' side of a market to find the price where the market settles, and shows what moves that price.

YOU WILL NEED

Chapter 3 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'; Chapter 6 'Consumption and Consumers' Surplus', section 6.2 'Consumers' Surplus'; Appendix A1 'Maths You Will Use'.

A shopper's choice depends on taste and on the size of her purse.

This chapter answers both questions with two lines on one graph.

The first line shows what pleases the shopper; the second shows what she can afford.

Suppose you enter a market stall that sells only two goods, mangoes and guavas, and you have Tk 100 to spend. You could buy only mangoes. You could buy only guavas. Or you could buy some of each, in many different mixes. Every mix may cost the same, yet you will not feel the same about them. Which mix should you choose? Most people say, "The one I like best." That answer is true, but it hides two questions. What does it mean to like one mix as much as another? And how does the purse limit the choice?

Chapter 3 gave one way to think about taste: count the satisfaction, or utility, of each mix. Counting is difficult, because nobody can look inside a buyer's mind and read a number. This chapter offers a second way that needs no counting. It asks only that a shopper can say whether one mix is better than, worse than, or equal to another. From that single ability we can draw a picture of her taste, and then add a picture of her purse.

The picture has two parts. The first is a line of taste: all the mixes that please the shopper equally. The second is a line of limit: all the mixes that her money can buy. Where the two lines meet in the right way, we shall find her best choice. Section by section, we give each line its proper name, learn how to measure its steepness, and then bring the two together. Many students find a graph with several lines confusing at first, because each line answers a different question. We therefore go one line at a time.

* * *

7.1 INDIFFERENCE CURVES

Begin with an everyday case. A friend offers you a plate of fruit. Plate one holds 4 mangoes and 2 guavas. Plate two holds 3 mangoes and 4 guavas. After a moment's thought you say that you would be equally happy with either plate. On plate two you have lost one mango and gained two guavas, and your pleasure has not changed. In ordinary speech you are *indifferent* between the plates, which means you do not mind which one you receive.

Economists use the same idea with two goods of their choosing. Let the goods be notebooks and pens. A particular amount of each good, such as 4 notebooks and 14 pens, is called a *combination*. (Some books say *bundle*; this chapter uses *combination* throughout.) For any two combinations, a consumer can say: this one is better, that one is better, or they are equal. That is the only ability we need, and every adult shopper has it.

Now you can name the main tool of the chapter. An *indifference* curve is a line that shows all the combinations of two goods that give the same level of utility. Recall: utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'.

GO SLOWLY HERE.

A friend's two plates of fruit can please you equally.

Economists call a particular amount of each of two goods a combination.

Indifference curves are lines that show all the combinations of two goods that give the same level of utility.

REVIEW QUESTIONS

- 8.1 Plot the supply schedule of the notebook from the case file on a graph, with price on the vertical axis. Say what the slope of the line shows.
- 8.2 After the tax on paper, the quantities supplied at Tk 20, 30, 40, 50 and 60 are 10, 20, 30, 40 and 50. Plot the new curve on the same graph as the old one. Then draw a third curve that shows an increase in supply of 15 at every price, and give its quantity at Tk 40.
- 8.3 Sort each cause as a price cause (movement along the curve) or a non-price cause (shift of the curve): (a) the notebook price rises from Tk 30 to Tk 40; (b) paper becomes cheaper; (c) a subsidy is paid to notebook makers; (d) the wholesaler lowers the notebook price from Tk 60 to Tk 50.
- 8.4 Using $Q_s = P$, find the quantity supplied at Tk 55 and the price at which 25 notebooks are supplied. Then, using $Q_s = P - 10$, find the quantity supplied at Tk 55.
- 8.5 State the law of supply, and name one condition that must stay the same for it to hold.
- 8.6 Name two kinds of exceptional supply curve, and give one example of each.
- 8.7 A new wholesaler enters the market, and the notebook's price is unchanged. Say what happens to supply, and name the determinant that changed.

Seven questions that check the chapter's ideas.

Indifference map. A set of indifference curves, with higher curves showing greater satisfaction.

Many curves together, higher ones better, form an indifference map.

Four properties of the curves follow from one assumption.

Recap of section 7.1.

Marginal rate of substitution. The amount of one good a consumer will give up to gain one more unit of another good.

GO SLOWLY HERE.

Moving along a curve means swapping one good for the other.

A worked example turns the swap into a number.

new combination would be better, so it could not lie on the same curve. This argument uses one assumption: the student always prefers more of a good to less.

PAUSE AND CHECK 7.P1 Combination P has 5 notebooks and 10 pens. Combination Q has 5 notebooks and 12 pens. Can P and Q lie on the same indifference curve? Give your reason.

One curve shows one level of satisfaction, but the student can imagine many levels, so there are many curves. An **indifference map** is a set of indifference curves, with higher curves showing greater satisfaction. Think of the contour lines on a walker's map, where each line joins places of equal height; here each line joins combinations of equal satisfaction, and a curve further from the corner of the axes is a higher hill. The student prefers 6 notebooks and 14 pens to 6 notebooks and 8 pens, because it has the same notebooks and more pens, so it lies on a higher curve.

Four properties follow from the assumption that the consumer prefers more of a good to less. First, each curve slopes downward, as the table showed. Second, a curve further from the corner where the axes meet gives greater satisfaction, because it holds more of the goods. Third, curves bend in towards that corner; section 7.2 'Marginal Rate of Substitution' explains why. Fourth, two curves never cross. The assumption has limits. If more of a good makes the consumer worse off, such as a sweet she has already eaten too many of, the first property fails. This chapter keeps to the usual case.

Before moving on. An indifference curve joins the combinations of two goods that please a consumer equally. It slopes down, and a map of such curves ranks every combination, with higher curves better and no two curves crossing. *You should now be able to say* what one point on a curve shows and why a higher curve is preferred.

7.2 MARGINAL RATE OF SUBSTITUTION

Return to the table of equally satisfying combinations. Moving from B (4 notebooks, 14 pens) to C (6 notebooks, 8 pens), the student gains 2 notebooks and gives up 6 pens, and her satisfaction stays the same. At what rate does she swap pens for notebooks? The **marginal rate of substitution** is the amount of one good a consumer will give up to gain one more unit of another good. Here she gives up 3 pens for each extra notebook, so the rate over this step is 3. The word *marginal* means "one more", as in the marginal utility of Chapter 3.²

Given. Combination B has 4 notebooks and 14 pens; combination C has 6 notebooks and 8 pens; both lie on one curve. *Find.* The rate at which pens are swapped for notebooks from B to C. *Formula.* In words: pens given up divided by notebooks gained. In symbols: rate = $F \div G$, where F is the fall in pens and G is the gain in notebooks. *Steps.* $F = 14 - 8 = 6$ pens. $G = 6 - 4 = 2$ notebooks. Rate = $6 \div 2 = 3$ pens per notebook. *Answer.* She gives up 3 pens for each

² Recall: marginal utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'.

The second case is fixed supply. On a market day, a fishmonger who has caught a fixed number of fish cannot offer more, however high the price rises. A single famous painting, and a fixed number of seats in a hall, are other examples. The curve is a vertical line: the quantity supplied is the same at every price. In the short period before more can be made, the price cannot change what is on offer.

Before moving on. An exceptional supply curve does not slope upward. A backward-bending curve comes from a seller who chooses less as the price rises, and a fixed supply comes from a quantity that cannot change in the short period. Both are limits of the law of supply.

WORKED CASE: USING THE WHOLESALE'S SUPPLY SCHEDULE

The case file gives a rule: the quantity supplied equals the price in taka. In symbols, $Q_s = P$, where Q_s is the quantity supplied in notebooks a week and P is the price in taka. After the tax on paper, the quantity supplied is 10 lower at every price, so the rule becomes $Q_s = P - 10$. Rearranged, the new rule gives $P = Q_s + 10$.⁴

Example 1. *Given:* $Q_s = P$. *Find:* the quantity supplied when the price is Tk 35, and the price at which 45 notebooks are supplied. *Formula:* quantity supplied equals the price in taka. *Steps:* at Tk 35, $Q_s = 35$; for 45 notebooks, $P = 45$. *Answer:* at Tk 35 the wholesaler supplies 35 notebooks a week, and 45 notebooks a week are supplied at Tk 45. *Check:* both answers fall between rows of the table, and the quantity rises with the price, as the law of supply says.

Example 2. *Given:* the old rule $Q_s = P$ and the new rule $Q_s = P - 10$ after the tax on paper. *Find:* the quantity supplied at Tk 50 under each rule, and the price at which 30 notebooks are supplied after the tax. *Formula:* $Q_s = P$ for the old rule, $Q_s = P - 10$ for the new rule, and $P = Q_s + 10$ for the price under the new rule. *Steps:* old rule, $Q_s = 50$; new rule, $Q_s = 50 - 10 = 40$; for 30 notebooks under the new rule, $P = 30 + 10 = \text{Tk } 40$. *Answer:* at Tk 50 the quantity falls from 50 to 40 notebooks a week, and the wholesaler needs a price of Tk 40 to offer 30 notebooks. *Check:* the table after the tax shows 30 notebooks at Tk 40, and the fall of 10 matches the case file.

Worked classification. *Situation:* in one month, six events happen. (a) The wholesaler raises the notebook price from Tk 40 to Tk 50. (b) A tax on paper is introduced. (c) A faster printing machine is installed. (d) A second wholesaler begins to sell the notebook. (e) Paper becomes dearer for another reason. (f) The notebook price falls from Tk 60 to Tk 50. *Question:* which events are movements along the supply curve and which are shifts? *Reasoning:* in (a) and (f) the price of the good itself changed. In (b) the tax changed, in (c) the technology changed, in (d) the number of sellers changed, and in (e) the cost of inputs changed. *Conclusion:* (a) and (f) are movements along the curve, and (b) to (e)

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

A fixed supply stays the same at every price.

The section closes with the two exceptional cases and their reasons.

Two links let us calculate any quantity or price from the supply rule.

The first example is a clean calculation of a quantity and a price.

The second example follows the shift caused by the tax.

Sorting events as price causes or non-price causes needs only one question.

Income shifts the line sideways; a price change turns it.

Recap of section 7.3.

Both pictures on one graph show the best affordable combination.

Consumer equilibrium. The point where a consumer gets the greatest satisfaction possible from a given income.

Consumer equilibrium is the best affordable combination.

The best point is where one curve touches the line and does not cross it.

notebook always costs the same 2 pens. Its steepness is the price ratio, 2. Compare this with the curve: the curve bends because her wishes change as her holdings change, while the line is straight because prices stay fixed.

Two changes alter the line. If income rises and prices stay the same, the whole line shifts outwards, parallel to the old one, because the price ratio is unchanged: with Tk 600 she could buy 15 notebooks or 30 pens. If one price changes, the line turns around the point on the axis of the other good. If pens fell to Tk 10, she could still buy only 10 notebooks, but now 40 pens, so the line turns outwards around the notebook point and becomes steeper.

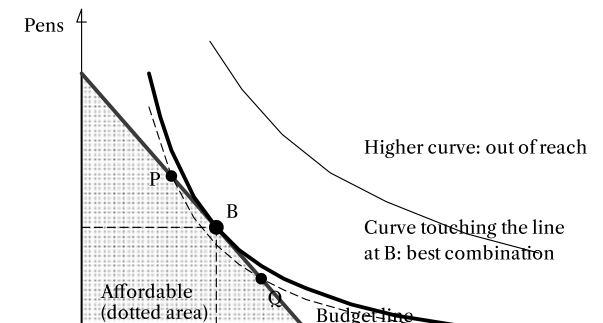
Before moving on. The budget line shows every combination that a given income buys at given prices. Its ends are income divided by each price, and its steepness is the price ratio. A rise in income shifts it; a price change turns it. *You should now be able to say where the line meets each axis and why it is straight.*

7.4 CONSUMER EQUILIBRIUM

We now hold two pictures. A map of curves shows what the student wants, and a budget line shows what she can afford. Draw both on one graph. Her aim is to reach the highest curve possible, and the budget line stops her from reaching any curve that lies beyond it. So the question becomes: which is the highest curve that still touches the budget line?

Consumer equilibrium is the point where a consumer gets the greatest satisfaction possible from a given income. The word *equilibrium* means a state of balance: once the consumer is there, she has no wish to change. For the student, it is the combination on the budget line that lies on the highest reachable curve.

In the figure below, the straight falling line is the budget line, and the two bent lines are indifference curves. The higher curve never meets the budget line, so the student cannot reach it. The lower curve crosses the line at two points, P and Q, and between them the line runs through combinations that lie on higher curves, so she can do better by moving. The best curve is the one that touches the line at a single point, marked B, without crossing. At B the curve and the line have the same steepness.



to encourage them to produce, and it lowers the seller's expense, so supply rises. The tax on paper in our opening example is a tax on an input.

Number of sellers. If a second wholesaler begins to sell the notebook, the quantity offered to the market at each price rises. If a wholesaler leaves, the quantity falls. The table below lists the four determinants. We introduce it in words first: each row names a determinant, and the last column gives the effect on supply.

THE DETERMINANTS OF SUPPLY, WITH A NOTEBOOK EXAMPLE

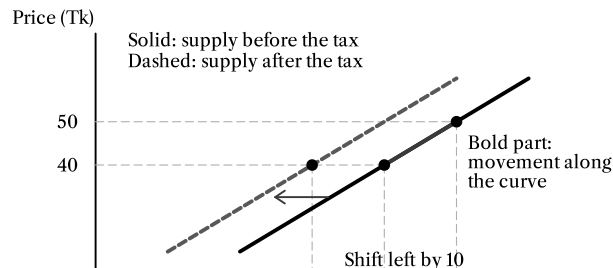
DETERMINANT	EXAMPLE FOR THE NOTEBOOK	EFFECT ON SUPPLY
Cost of inputs	Paper becomes dearer	Falls
Technology	A faster printing machine is installed	Rises
Taxes and subsidies	A tax on paper is introduced	Falls
Taxes and subsidies	A subsidy on paper is introduced	Rises
Number of sellers	A new wholesaler enters the market	Rises

To read a row, take the third line: when a tax on paper is introduced, the wholesaler's expense rises, and supply falls. The word *falls* here means that fewer notebooks are offered at every price, and not only at one price.

When a determinant changes, the whole curve moves to a new position. We call this a shift of the supply curve, which some books call a change in supply. This book uses the word shift. The case file gives a precise example.

Case file, illustrative. A tax on paper lowers the wholesaler's weekly supply of the notebook by 10 at every price. At a price of Tk 20 the quantity supplied becomes 10; at Tk 30, 20; at Tk 40, 30; at Tk 50, 40; at Tk 60, 50.

Now we draw the numbers, and this is the first chart of the chapter. Read it as follows. The vertical axis shows the price in taka, and the horizontal axis shows the quantity supplied in notebooks a week. A point means: at this price, the wholesaler offers this quantity. The solid line is the original supply curve, and the dashed line is the supply curve after the tax. The bold part of the solid line shows a movement along the curve, from Tk 40 to Tk 50.



More sellers in the market mean a larger quantity supplied at each price.

Read each row as a change of condition and the resulting change of supply.

A change in a determinant shifts the whole supply curve.

The case file records the shift caused by a tax on paper.

The chart shows the old curve, the new curve and a movement along the old curve.

Notebooks: $540 \div 60 = 9$. Pens: $540 \div 45 = 12$. Steepness: $60 \div 45 = 1.33$ pens per notebook. *Answer.* At most 9 notebooks or 12 pens, with 1.33 pens given up per notebook. *Check.* $9 \times 60 = \text{Tk } 540$ and $12 \times 45 = \text{Tk } 540$.

Four common slips spoil these calculations.

Recap of the worked case.

Common slips deserve a warning. Some students divide income by the wrong price, which swaps the two ends of the line. Some put the pen price on top of the price ratio, which turns the steepness upside down. Some forget the units, so that 1.33 loses its meaning of pens per notebook. Others test only one neighbour and declare a winner; test both sides of the touching point.

Before moving on. To find the best combination, compute the ends of the budget line, find the price ratio, locate the point where the rate equals it, and test the neighbours. *You should now be able to say* how each of the five steps supports the answer.

* * *

SUMMARY

An indifference curve joins combinations that please a consumer equally, and a map of such curves ranks every combination. The marginal rate of substitution measures the pens she will swap for one more notebook, and it falls along a curve. The budget line shows what her income buys at given prices, and its steepness is the price ratio. Consumer equilibrium is where the budget line touches the highest curve it can reach; there the rate equals the price ratio.

THINK IT THROUGH

7.T1 Income is Tk 400, notebooks cost Tk 40, and pens fall in price from Tk 20 to Tk 10. Show that the student can still afford 6 notebooks and 8 pens. What does that tell you about her satisfaction after the price fall?

7.T2 Explain why two indifference curves cannot cross. Use the idea that the consumer prefers more of a good to less.

WHERE PEOPLE GO WRONG

Some students think that being indifferent means not caring. The student cares a great deal. She is indifferent only between combinations on the same curve, and she strongly prefers any combination on a higher curve to one on a lower curve.

Some students read an indifference curve as a list of what the consumer can buy. It is not. The curve says nothing about money. Only the budget line shows affordability, and a point on a curve may cost more than her income.

Some students treat the marginal rate of substitution and the price ratio as one idea. The first comes from her wishes and changes along a curve. The second comes from the market and stays fixed along a budget line. They are equal only at consumer

Indifferent does not mean uninterested.

The curve shows taste, not what she can afford.

The rate and the price ratio are different things.

The simplest record of supply is a table. A **supply schedule** is a table showing the quantity supplied at each price. We now use the case of the shop's own wholesaler. The next paragraph is a record, and its figures are fixed.

Case file, illustrative. The wholesaler who sells the 200-page notebook to Meghna and Sons records its weekly supply. At a price of Tk 20 the quantity supplied is 20 notebooks; at Tk 30, 30; at Tk 40, 40; at Tk 50, 50; at Tk 60, 60. The quantity supplied equals the price in taka.

WEEKLY SUPPLY SCHEDULE FOR THE 200-PAGE NOTEBOOK

PRICE (TK)	QUANTITY SUPPLIED (NOTEBOOKS A WEEK)
20	20
30	30
40	40
50	50
60	60

To read a row, take the line for Tk 40. It says that at a price of Tk 40, the wholesaler chooses to offer 40 notebooks a week. Now read down the table. As the price rises by Tk 10 each time, the quantity supplied also rises by 10 each time. Compare this with the demand schedule of Chapter 4: there the quantity fell as the price rose, and here it rises.¹ The rule in the case file, quantity equals price, gives the same numbers.

The pattern has a name. The **law of supply** says that when the price of a good rises, the quantity supplied rises, if other conditions stay the same. The law works in both directions: when the price falls, the quantity supplied falls. As in the law of demand, the final clause matters. Costs, technology and taxes must stay fixed, or we cannot tell whether a change in the quantity offered came from the price.

Why do sellers behave in this way? Two reasons are enough for now. The first is the reward. A higher price makes each extra unit more profitable to offer, so a seller will use more hours, more materials and more space. The second is the expense of expanding. Making a large quantity usually costs more for each extra unit than making a small quantity, because the best fields, the best workers and the best machines are used first. A seller will accept the higher expense of the extra units only if the price is high enough to pay for them.

Notice how the word *cost* is used in the paragraph above. This book gives it a narrow meaning: the expense that a seller meets in making and delivering a good. What the buyer pays is called the price. A notebook may have a price of Tk 40 and cost the wholesaler less than that to supply. Keeping the two words apart avoids many confusions later.

Supply schedule. A table showing the quantity supplied at each price.

A supply schedule is a table of prices and the quantities supplied at them.

The case file records the weekly supply of the notebook by its wholesaler.

The same record is laid out as a table.

Each row pairs one price with one quantity supplied.

Law of supply. When the price of a good rises, the quantity supplied rises, if other conditions stay the same.

The pattern in the table is the law of supply.

Two reasons explain why a higher price brings a larger quantity supplied.

Cost. Word in use. In daily life, cost often means what a buyer pays for something. In economics, cost means the seller's expense in making the good, and what the buyer pays is the price.

In this book, cost means the seller's expense, and price

¹ Recall: demand, Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'.

Before such mills, much cotton yarn was spun by hand on wheels in workers' homes. Arkwright gathered the spinning into one building, where a water wheel and machines did work that people had done by hand. In the language of this chapter, the mix moved toward machines and power. He also needed workers at the site, and in 1776 he began building houses for them in Cromford. We do not know what calculations he made, but his question was the one of this chapter: which mix of workers, machines and power makes the yarn at the lowest cost?

The mill moved spinning from workers' homes to machines, and workers needed housing at the new site.

REVIEW QUESTIONS

12.1 Define an isoquant and an isocost line in your own words, and state one difference between an isoquant and an indifference curve.

12.2 With a wage of Tk 600 and a machine rent of Tk 400, draw the isocost line for a total of Tk 4,800 by finding where it meets each axis. State its slope.

12.3 A firm must make 10 units. The mixes of (workers, machines) that give this output are (1, 8), (2, 4), (4, 2) and (8, 1). A worker costs Tk 30 a day and a machine costs Tk 10 a day. Find the least-cost combination.

12.4 Use the case mixes for 100 books and the wage of Tk 600, but let the machine rent fall from Tk 400 to Tk 300. Find the least-cost combination, and compare the MRTS between the cheapest mixes with the new price ratio.

12.5 In the case, find the wage bill as a percentage of total cost at the least-cost combination. Then find it again after the wage rises to Tk 900 and the firm moves to its new best mix. Explain the change.

12.6 The marginal product of a worker is 30 books and that of a machine is 10 books. The wage is Tk 600 and the machine rent is Tk 400. Which input adds more output per taka, and how should the firm change its mix?

Worked example 2 combines a market with two curves and a shift.

The steps of worked example 2, before and after the shift.

Common slips: units, signs and mixed-up shifts cause most wrong answers.

The whole chapter in a few lines.

Two short problems apply the chapter to real choices.

(5) Average $P = (200 + 240) \div 2 = \text{Tk } 220$. (6) $40 \div 220 = 0.1818$. (7) $E_s = 0.0952 \div 0.1818 = 0.52$. *Answer.* The elasticity of supply is about 0.52, so supply is inelastic. *Check.* The quantity rose by about 9.5% and the price by about 18.2%. A smaller percentage over a larger one gives a value below 1, as found.

The second example joins the earlier techniques. It uses invented figures for a market in rice sacks. First we find the equilibrium; then demand shifts, and we find it again.

Given. Quantity demanded = $70 - \text{price}$. Quantity supplied = $2 \times \text{price} - 20$. Later, demand shifts to a quantity demanded of $100 - \text{price}$. *Find.* The equilibrium price and quantity before and after the shift. *Formula.* At equilibrium, quantity demanded = quantity supplied. *Steps.* Before: (1) $70 - \text{price} = 2 \times \text{price} - 20$. (2) Add 20 to both sides: $90 - \text{price} = 2 \times \text{price}$. (3) Add price to both sides: $90 = 3 \times \text{price}$. (4) $\text{Price} = 90 \div 3 = 30$. (5) Quantity = $70 - 30 = 40$. After: (6) $100 - \text{price} = 2 \times \text{price} - 20$. (7) $120 = 3 \times \text{price}$. (8) $\text{Price} = 40$. (9) Quantity = $100 - 40 = 60$. *Answer.* The equilibrium moves from a price of 30 and a quantity of 40 to a price of 40 and a quantity of 60. *Check.* At a price of 40, quantity demanded is $100 - 40 = 60$ and quantity supplied is $2 \times 40 - 20 = 60$. Demand shifted right, and both price and quantity rose, as the table says.

Three slips cause most wrong answers. First, forgetting to divide by the average and using the start value only. Second, writing the ratio upside down: the percentage change in quantity goes on top. Third, calling a price change a shift. Elasticity has no unit, because it divides one percentage by another; do not write taka or notebooks after it.

* * *

SUMMARY

Elasticity of supply is the percentage change in quantity supplied divided by the percentage change in price, found with the mid-point method. Values above 1 are elastic and below 1 are inelastic. In a competitive market, the equilibrium price makes the quantity demanded equal the quantity supplied. Excess demand pushes the price up; excess supply pushes it down. A shift of demand or supply moves the equilibrium to a new price and quantity.

THINK IT THROUGH

Use the notebook market of this chapter to answer the first question. Use the factors of section 9.1 for the second.

9.T1 A law forbids any price above Tk 30 for a notebook. Using the notebook market, state the quantity demanded, the quantity supplied and the excess demand at Tk 30, and say who gains and who loses.

9.T2 A farmer says, "Give me a year and I can supply far more rice, but this month I cannot." Which factor from section 9.1 does she describe, and what does it say about the elasticity of supply this month and next year?

same output. How large the effect is depends on how easily machines replace workers, which is the curvature of the isoquant.

Before moving on. Repeating the method for each output gives the expansion path. Minimising cost for a given output is not the same as maximising profit. In the short run a fixed input can prevent the firm from reaching the least-cost combination.

* * *

WORKED EXAMPLES: FINDING THE LEAST-COST MIX

We now apply the rule in full, in the order Given, Find, Formula, Steps, Answer and Check. The first example uses small numbers, so that the method is the only new thing to learn. The second uses the case and a change in the wage.

Example 1: output per taka. Given. The wage is Tk 20 per worker per day and the machine rent is Tk 10 per machine per day. At the firm's present mix, the marginal product of a worker is 40 books and that of a machine is 15 books. *Find.* Whether the mix is the least-cost combination and, if not, which way the firm should change it.

Formula. In words: the marginal product per taka equals the marginal product divided by the price of the input. The mix is least-cost when the ratio is equal for both inputs. In symbols: $MP \text{ of a worker} / w = MP \text{ of a machine} / r$, where MP means marginal product, w is the wage and r is the machine rent. *Steps.* Workers: $40 / 20 = 2$ books per taka. Machines: $15 / 10 = 1.5$ books per taka. Compare: 2 is greater than 1.5.

Answer. The mix is not least-cost. A taka spent on workers adds more output than a taka spent on machines, so the firm should hire more workers and rent fewer machines. *Check.* The MRTS is $40 / 15 = 2.67$, which is above the price ratio $20 / 10 = 2$. The rule of section 12.2 also says to hire more workers. The two tests agree.

Example 2: a wage rise. Given. The output is 100 books a day. The machine rent stays at Tk 400. The wage rises from Tk 600 to Tk 900. The mixes on the isoquant are (2, 10), (3, 6), (4, 4), (5, 3) and (6, 2). *Find.* The new least-cost combination and the rise in total cost.

Formula. $C = w \times L + r \times M$, with $w = 900$ and $r = 400$. *Steps.* (2, 10): $900 \times 2 + 400 \times 10 = 1,800 + 4,000 = \text{Tk } 5,800$. (3, 6): $2,700 + 2,400 = \text{Tk } 5,100$. (4, 4): $3,600 + 1,600 = \text{Tk } 5,200$. (5, 3): $4,500 + 1,200 = \text{Tk } 5,700$. (6, 2): $5,400 + 800 = \text{Tk } 6,200$. The smallest is Tk 5,100. The rise is $5,100 - 4,000 = \text{Tk } 1,100$.

Answer. The new least-cost combination is 3 workers and 6 machines, at Tk 5,100. Total cost rises by Tk 1,100. *Check.* The price ratio is now $900 / 400 = 2.25$. The MRTS between (2, 10) and (3, 6) is 4, which is above 2.25, so the firm moves. Between (3, 6) and (4, 4) it is 2, which is below 2.25, so the firm stops at (3, 6). Keeping the old mix (4, 4) would cost Tk 5,200, a rise of Tk 1,200, so changing the mix saves Tk 100.

Three slips are common. The first is to reverse the MRTS: divide

Recap: the expansion path, the two kinds of choice and the short run.

Two examples apply the rule: the first with small numbers, the second with case data and a wage rise.

Example 1: test a mix by comparing output per taka from each input.

The test divides each marginal product by the price of its input.

Workers give more output per taka, so the firm should hire more workers and fewer machines.

Example 2: the wage rises from Tk 600 to Tk 900 in the case.

Total cost is recomputed for each mix at the new wage and the smallest is chosen.

The best mix moves to 3 workers and 6 machines, and the rise is Tk 1,100.

Three slips are common:

COST OF EACH MIX THAT GIVES 100 BOOKS A DAY

WORKERS, MACHINES	WAGE BILL (TK)	RENT BILL (TK)	TOTAL COST (TK)
2, 10	1,200	4,000	5,200
3, 6	1,800	2,400	4,200
4, 4	2,400	1,600	4,000
5, 3	3,000	1,200	4,200
6, 2	3,600	800	4,400

Read the last column from top to bottom. The total falls from 5,200 to 4,000 and then rises to 4,400. The lowest value is the least-cost combination, 4 workers and 4 machines. The cheapest mix is neither the one with most workers nor the one with most machines. The best mix lies in the middle, where the two bills are balanced.

Now we draw the same facts. On the chart, the horizontal axis shows the number of workers and the vertical axis shows the number of machines. A point is a mix. The curved line is the isoquant for 100 books. The straight lines are isocost lines, and a line farther from the corner means a higher total cost. The firm wants the lowest isocost line that still touches the isoquant. That line touches the curve at one point, without crossing it, which is called a tangent. The point of contact is the least-cost combination.

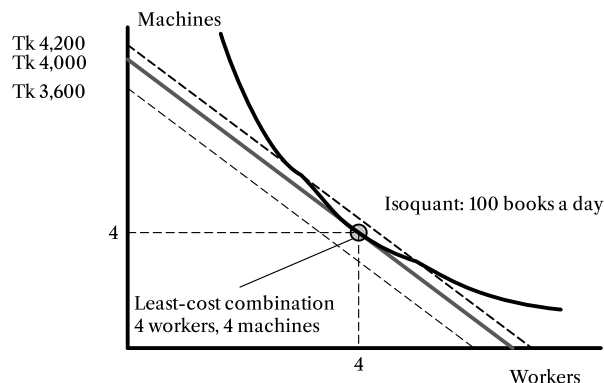


FIG. 12.1. Isoquant for 100 books with isocost lines, and the least-cost combination

At the point of contact the two lines have the same slope. The slope of the isoquant is the MRTS, and the slope of the isocost line is the price ratio. So the least-cost rule is: $MRTS = \text{wage} / \text{machine rent}$. In the case, $600 / 400 = 1.5$. Between (3, 6) and (4, 4) the MRTS is 2, which is above 1.5. Between (4, 4) and (5, 3) it is 1, which is below 1.5. The ratio 1.5 lies between them, at (4, 4).

The reason can be shown with numbers. At (3, 6), one more worker saves two machines worth Tk 800 and costs Tk 600, so the firm gains Tk 200. It should move. At (4, 4), one more worker saves one machine worth Tk 400 and costs Tk 600, so the firm loses Tk 200. It should stop. The firm adjusts until the gain from changing

Total cost falls and then rises, and the lowest point is the least-cost combination.

On the chart, the cheapest isocost line that touches the isoquant shows the best mix.

YOU WILL NEED

Chapter 2, 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost', for the idea of giving up one use to gain another; Chapter 8, 'Supply and the Supply Curve', section 8.1 'Quantity Supplied and the Law of Supply', for supply; and Appendix A1, 'Maths You Will Use', for ratios, averages and simple graphs.

A workshop owner adds workers one by one and watches the output.

By the sixth worker the bench is crowded and the output no longer grows.

The puzzle is why extra workers first add more and then add less.

A bookbinder owns a small workshop in an old street. It has a single long bench and one sewing machine. On Monday the owner hires one worker, who folds the pages, sews them and fits the covers alone, and the day's output is modest. On Tuesday a second worker arrives. One folds while the other sews, and the output is more than double. Encouraged, the owner hires a third worker, then a fourth. The output keeps rising, but each new worker adds a little less than the last.

By the sixth day the bench is crowded. Workers wait for the machine, carry the same pile of paper from hand to hand and step round one another. The sixth worker is as willing as the first, and works as hard, yet the day's output does not rise at all. The owner has paid for one more worker and received nothing for it. Nothing about the workers has changed. Something about the bench has.

This is the puzzle of the chapter. Why did the second worker add more than the first, and why did the sixth add nothing? If extra workers always added as much as the first, one workshop could feed the whole city, and we would never need a second bench. Section 10.1 names the parts of the problem and defines the measures. Section 10.2 explains the pattern, which is among the most important in economics. Section 10.3 asks what happens when the owner builds a bigger workshop and raises every input at once.

* * *

10.1 INPUTS AND OUTPUT

THE RULE THAT LINKS THEM

Begin in a kitchen. A baker uses flour, water, an oven, fuel and time to make loaves. The things used are the inputs, and the loaves are the output. Inputs are the workers, machines, materials and land that a firm uses in production; output is the quantity of goods that results. A baker with more flour and a second oven can make more loaves, but not without limit, and not in any proportion.

Every input has another use, so using it has an opportunity cost, as Chapter 2 explained. A firm therefore wants the largest output from the inputs it has chosen. Engineers and owners can describe the relationship between inputs and output in a table, a graph or a rule. Economists give the rule a name.

A **production function** means a rule showing the largest output that can be made from each combination of inputs. Suppose a baker with 1 oven and 2 helpers can make at most 100 loaves a day.

GO SLOWLY HERE.

A baker's kitchen shows how inputs become output.

Recall: opportunity cost, Chapter 2, section 2.2 'Opportunity Cost'.

Production function. A rule showing the largest output that can be made

At the point of contact the slope of the isoquant equals the slope of the isocost line.

If the MRTS is above the price ratio, the firm should hire more workers and fewer machines.

The line has a name. An **isoquant** is a line showing all combinations of two inputs that produce the same output. The word *iso* means equal, and *quant* comes from quantity, so isoquant means equal quantity of output. If five mixes each give 100 books a day, all five lie on one isoquant.

Case file, illustrative. For an output of 100 books bound a day, the wage is Tk 600 per worker per day and the machine rent is Tk 400 per machine per day. Five combinations give this output, written as (workers, machines): (2, 10), (3, 6), (4, 4), (5, 3) and (6, 2). Joined together, they form one isoquant.

An isoquant has three properties. First, it slopes downward: to keep output the same, more workers mean fewer machines. Second, it curves toward the corner where the two axes meet, because each added worker replaces fewer machines than the one before. Third, two isoquants never cross, and an isoquant farther from the corner shows a larger output. These properties match those of an indifference curve and an indifference map, with one difference. Satisfaction cannot be counted, but output can be counted in books.

Now consider the owner's money. Suppose the owner decides to spend Tk 4,000 a day on inputs. With that sum the owner can hire many workers and rent few machines, or the reverse. Every division of Tk 4,000 between the two inputs lies on one line.

An **isocost line** is a line showing all combinations of two inputs that cost the same total amount. Here *cost* is the firm's expense. The formula in words is: total cost equals the wage times the number of workers, plus the machine rent times the number of machines. In symbols, $C = w \times L + r \times M$, where C is the total cost, w is the wage per worker, L is the number of workers, r is the rent per machine and M is the number of machines.

Take the case prices. If all of Tk 4,000 buys workers, the firm hires $4,000 / 600 = 6.67$ workers a day, which is an average over several days. If all of it buys machines, the firm rents $4,000 / 400 = 10$ machines. These are the two ends of the line. The mix (4, 4) lies on it, because $600 \times 4 + 400 \times 4 = 2,400 + 1,600 = \text{Tk } 4,000$. A higher total cost gives a line farther from the corner.

The slope of the line depends only on the two prices. One worker costs Tk 600 and one machine costs Tk 400, so a worker costs as much as $600 / 400 = 1.5$ machines. Along the line, each extra worker must be paid for by renting 1.5 fewer machines. If the wage rises, the line becomes steeper. If the machine rent rises, it becomes flatter.

PAUSE AND CHECK 12.P1 An isocost line is drawn for a total of Tk 4,800, with a wage of Tk 600 and a machine rent of Tk 400. How many workers could be hired if no machine is rented?

Chapter 7 gave the consumer's marginal rate of substitution, which keeps satisfaction the same. The firm has a twin, and it keeps output the same. The **marginal rate of technical substitution** is the amount of one input a firm can give up when it adds one unit of another, keeping output the same. We write it as MRTS. Here it is the number of machines a firm can save when it adds one

Isoquant. A line showing all combinations of two inputs that produce the same output.

An isoquant joins all mixes that give the same output.

Case file: five mixes of workers and machines give 100 books a day.

Isoquants slope downward, curve toward the origin, and never cross.

A firm decides how much to spend and can divide the sum between workers and machines.

Isocost line. A line showing all combinations of two inputs that cost the same total amount.

An isocost line joins all mixes that have the same total cost.

The Tk 4,000 line meets the axes at 6.67 workers and 10 machines.

The slope of an isocost line is the price ratio: one worker costs as much as 1.5 machines.

Marginal rate of technical substitution. The amount of one input a firm can give up when it adds one unit of another, keeping output the same.

Recall: marginal rate of substitution. Chapter 7.

and sound alike. Remember the questions: total product asks how much altogether, marginal product asks how much the last unit added, and average product asks how much each unit produced on average.

PAUSE AND CHECK 10.P1 A baker with one oven makes 5 loaves with 1 helper, 12 with 2 helpers and 18 with 3 helpers. Find the marginal product of the third helper and the average product with 3 helpers.

You should now be able to say what total, marginal and average product ask: how much altogether, how much the last unit added, and how much per unit on average.

* * *

10.2 THE LAW OF DIMINISHING RETURNS

THE PATTERN IN THE WORKSHOP

Return to the bookbinder. The workshop has one machine, and the owner varies only the number of workers. The record shows what happened.

Case file, illustrative. A binding workshop has one machine. With 1 to 6 workers, daily output is 10, 24, 36, 44, 48 and 48. The marginal product is 10, 14, 12, 8, 4 and 0. The average product is 10, 12, 11, 9.6 and 8.

To read the table below, go along each row. The first column gives the number of workers, the second gives total product, the third gives the marginal product of the last worker, and the fourth gives average product. Each marginal product is the difference between a total product and the one above it. Each average product is a total product divided by the number of workers on its row.

THE BINDING WORKSHOP'S PRODUCT SCHEDULE

WORKERS	TOTAL PRODUCT	MARGINAL PRODUCT	AVERAGE PRODUCT
1	10	10	10
2	24	14	12
3	36	12	12
4	44	8	11
5	48	4	9.6
6	48	0	8

Follow the marginal column. The first worker adds 10 and the second adds 14, because two people can share the tasks and keep the machine busy. The third adds 12, the fourth 8, the fifth 4 and the sixth 0. From the third worker onward every extra worker adds less than the one before, while total product still rises until the fifth worker. The sixth leaves total product at 48.

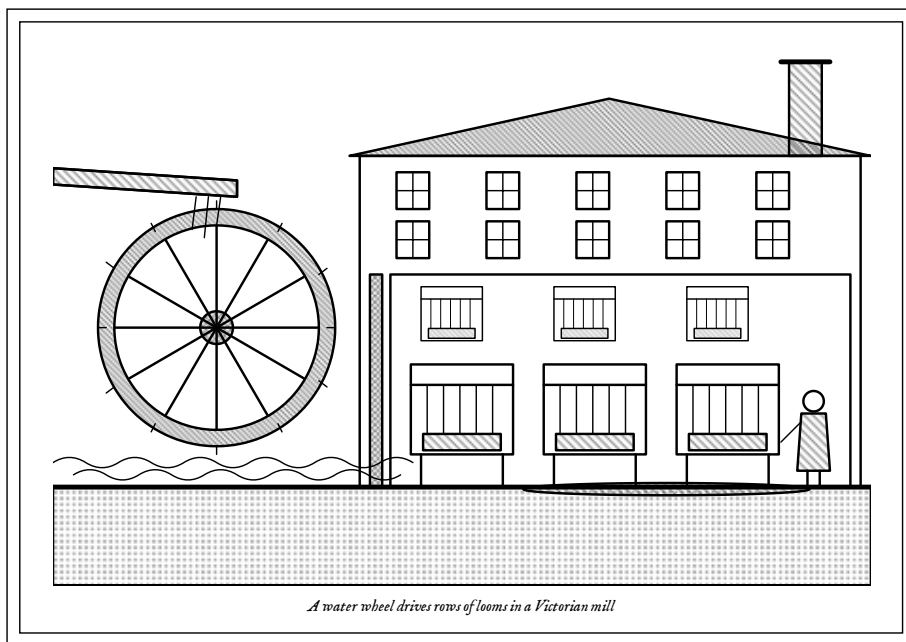
PAUSE AND CHECK 10.P2 In the schedule, what does the figure 8 in the marginal column mean, and what is the average product when 4 workers are used?

The binding workshop gives the pattern in numbers.

Recall: supply, Chapter 8, section 8.1 'Quantity Supplied and the Law of Supply'.

The schedule is a table that lists the three measures worker by worker.

Total product rises throughout, but by smaller and smaller steps after the second worker.



CHAPTER XII

The Least-Cost Combination of Inputs

How can a firm tell which mix of workers and machines is the cheapest way to make the same output?



This chapter shows how a firm chooses among the many ways of making the same output. You will learn to draw and read two lines, one for output and one for total cost, and to find the mix of inputs that costs least.

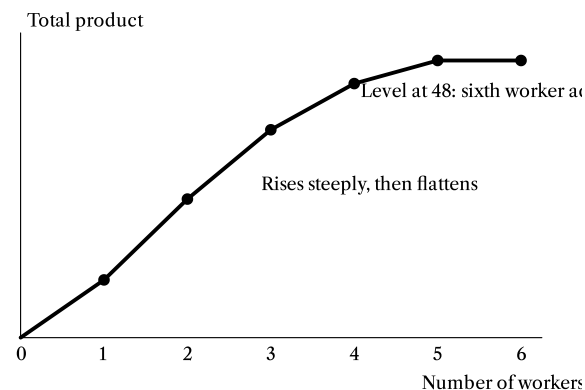


FIG. 10.1. *Total product curve for the binding workshop*

The second figure shows the marginal and average curves, which meet at the highest average.

The second figure draws the other two measures. The solid line is marginal product and the dashed line is average product. Marginal product climbs to 14 at the second worker, then falls to 0. Average product climbs to 12, stays there for the third worker and then falls. The curves meet at the third worker, where both equal 12, and this is where average product is at its highest.

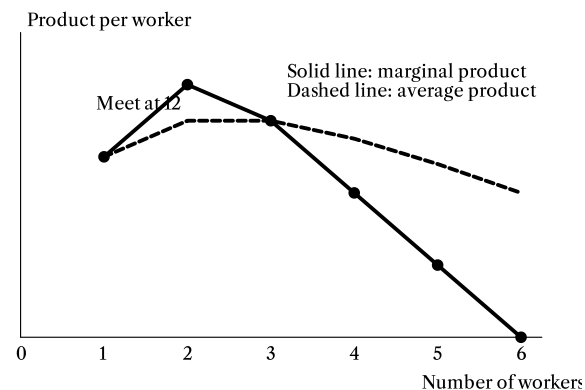


FIG. 10.2. *Marginal and average product curves for the binding workshop*

The marginal product pulls the average up or down.

The link between the two curves is a rule that deserves attention. When marginal product is above average product, the average rises; when it is below, the average falls; when the two are equal, the average neither rises nor falls. Check it against the table. The second worker adds 14, above the average of 10, so the average climbs to 12. The third adds 12, equal to the average, so the average stays at 12. The fourth adds 8, below 12, so the average falls to 11. The same rule governs the average mark of a class: a new student above the average raises it, and one below lowers it.

THREE STAGES OF PRODUCTION

* * *

SUMMARY

Costs are explicit, when money is paid to outsiders, or implicit, when the owner supplies the resource. In the short run, total cost is fixed cost plus variable cost. Marginal cost is the change in total cost divided by the change in output, and average total cost is total cost divided by output. Marginal cost falls and then rises, average total cost is U-shaped, and marginal cost crosses average total cost at its lowest point.

* * *

THINK IT THROUGH

11.T1 In the workshop table, variable cost per book rises from Tk 24 at 40 books to Tk 26 at 50 books, yet average total cost falls from Tk 39 to Tk 38. Explain how this can happen.

11.T2 A bookseller says, "My own time is free, because I pay myself nothing." Reply using the idea of implicit cost, and say what question she should ask.

* * *

WHERE PEOPLE GO WRONG

Mistake: "A fixed cost never changes." The cost is fixed only in the short run, and only as a total. In the long run every input can change, so no cost stays fixed. Fixed cost per book also changes, because the same total is shared among more books.

Mistake: "If nobody sends a bill, there is no cost." An implicit cost has no bill, but it has an opportunity cost. The owner's time and money could have been used elsewhere.

Mistake: "Marginal cost is total cost divided by output." That is average total cost. Marginal cost divides the change in total cost by the change in output. At 30 books, marginal cost is Tk 20 and average total cost is Tk 44.

Mistake: "Average cost is lowest where marginal cost is lowest." In the workshop, marginal cost is lowest at 30 books (Tk 20), but average total cost is lowest at 50 books (Tk 38). The average stops falling only when marginal cost rises to meet it.

* * *

IN HISTORY

In 1908, the Ford Motor Company of the United States began to sell the Model T, a car for ordinary families, at about USD 825. By 1925, the basic model cost about USD 260. In 1913, Ford introduced

The chapter's main points fit in a few lines.

A fixed cost is fixed only in the short run, and only as a total.

No bill does not mean no cost.

Marginal cost and average total cost answer different questions.

The two lowest points are at different outputs.

A famous fall in the price of a car shows unit cost falling as output grows.

decreasing. *Check:* the pattern of 2.5, 1.5 and 1.2 agrees with the pattern of inputs.

CAUSES

Why increasing returns at first? A larger workshop allows each worker to specialise, can use larger machines that a small workshop cannot afford, and can buy materials in bulk at lower prices. Why decreasing returns later? A very large firm is hard to manage. Orders pass through more people, mistakes spread, and effort is spent on coordination rather than production. Constant returns lie between: the firm repeats what it already does. Many firms pass through all three as they grow.

Do not mix up the two ideas. The table below compares them. Read each row across: the first column names the feature, and the next two give the answer for each idea.

DIMINISHING RETURNS COMPARED WITH RETURNS TO SCALE

FEATURE	LAW OF DIMINISHING RETURNS	RETURNS TO SCALE
Inputs changed	One input; others fixed	All inputs together
Period	Short run	Long run
Question	How does the extra output change?	How does output change with the size of the firm?
Result	Extra output shrinks	Output may rise by more, the same or less

Before moving on. Returns to scale compares the proportional rise in output with the proportional rise in all inputs. Increasing means output rises faster, constant means equally, and decreasing means more slowly. The law of diminishing returns belongs to the short run, and returns to scale to the long run.

* * *

WORKED EXAMPLES: FOUR RUNGS

Each example below follows the order Given, Find, Formula, Steps, Answer, Check.

Given: a baker with one oven makes 5, 12, 18 and 20 loaves with 1, 2, 3 and 4 helpers. **Find:** the marginal and average product at each number of helpers. **Formula:** $MP = \Delta TP \div \Delta L$ and $AP = TP \div L$. **Steps:** MP: 5; $12 - 5 = 7$; $18 - 12 = 6$; $20 - 18 = 2$. AP: $5 \div 1 = 5$; $12 \div 2 = 6$; $18 \div 3 = 6$; $20 \div 4 = 5$. **Answer:** MP is 5, 7, 6, 2 and AP is 5, 6, 6, 5. **Check:** the marginal product is above the average product while the average rises, and equals it at the peak of 6 with 3 helpers.

Given: on one acre, 1 to 5 labourers harvest 420, 900, 1,300, 1,560 and 1,690 kg of rice. **Find:** the marginal and average product. **Steps:** MP: 420; 480; 400; 260; 130. AP: 420; 450; 433.33; 390; 338. **Answer:** marginal product peaks with the second labourer and falls after, while average product peaks at the second as well. **Check:** the

Specialisation and bulk buying favour growth; coordination difficulties limit it.

The law of diminishing returns and returns to scale are different ideas.

Before moving on: the long run asks about all inputs, the short run about one.

Four examples, from clean numbers to a combined problem.

Rung 1: a table with small clean numbers.

Rung 2: realistic data in kilograms are less tidy.

Marginal cost therefore falls at first and then rises. In the table it falls from 30 to 20 and climbs to 48.

The curve of average total cost is shaped like the letter U. Two forces explain this. The first force pulls the average down: fixed cost is shared among more books as output grows, so average fixed cost falls. The second force pulls the average up: after a point, rising marginal cost makes average variable cost rise. At low output the first force is stronger, and the average falls. At high output the second force is stronger, and the average rises. Near 50 books the forces balance, and average total cost is at its lowest, Tk 38.

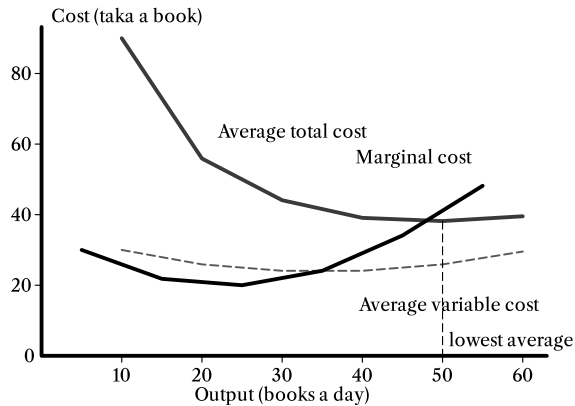


FIG. 11.2. *Marginal, average variable and average total cost curves of the binding workshop*

The second figure draws the three per-unit costs. The vertical axis now shows cost per book, not cost for the whole output. Look at where the marginal cost curve meets the average curves. Think of the average age of a class. A new student who is younger than the average lowers it, and a new student who is older raises it. The new student is the marginal addition. Costs follow the same rule. While the marginal cost of the next book is below average total cost, the average falls. When marginal cost is above it, the average rises. So marginal cost must cross average total cost at the bottom of the U.

The table confirms the rule. For the step from 40 to 50 books, marginal cost is Tk 34, below the average of Tk 39 at 40 books, so the average falls to Tk 38. For the step from 50 to 60 books, marginal cost is Tk 48, above the average of Tk 38, so the average rises to Tk 39.67. The same rule makes marginal cost cross average variable cost at its lowest point. Our table moves in steps of ten books, so the exact crossing lies between two rows.

So far one input, the machine, has been fixed. In the long run all inputs can be changed, so the firm has no fixed cost at all. It can choose a larger machine or a smaller one. The long-run cost curve shows the lowest average cost at which each output can be made when the firm is free to choose its size. It is usually also U-shaped, but for a different reason: the cause is returns to scale, taught in

Average total cost has a U shape because two forces pull on it in opposite directions.

Marginal cost crosses average total cost at the lowest point of the average.

The workshop figures show the crossing between two rows of the table.

In the long run every input can change, so the cost curve changes with it.

allowed. In the language of this chapter, improved methods moved the whole schedule upward.

REVIEW QUESTIONS

- 10.1 Define production function, and say why it concerns the largest output.
- 10.2 Distinguish the short run from the long run, and give an example of a fixed input.
- 10.3 A farm with fixed land has outputs of 30, 70, 100 and 120 kg with 1, 2, 3 and 4 labourers. Complete a table of marginal and average product.
- 10.4 In the binding workshop, explain why the average product stays at 12 for the third worker.
- 10.5 Inputs are multiplied by 3 and output is multiplied by 2.4. Classify the returns to scale.
- 10.6 Inputs are multiplied by 2 and output is multiplied by 2.6. Classify the returns to scale.
- 10.7 State two differences between the law of diminishing returns and returns to scale.

Managers decide one step at a time, so they ask about the next unit. **Marginal cost** is the extra cost of producing one more unit of output. In words, marginal cost equals the change in total cost divided by the change in output. In symbols, $MC = (\text{change in } TC) \div (\text{change in } Q)$, where MC is marginal cost and Q is the quantity of output. The tea stall's total cost rises from Tk 140 at 10 cups to Tk 144 at 11 cups. The change in TC is Tk 4 and the change in Q is 1 cup, so $MC = 4 \div 1 = \text{Tk } 4$.

The same rule works when output rises ten books at a time. In the workshop, total cost rises from Tk 900 at 10 books to Tk 1,120 at 20 books. The change in TC is Tk 220, and the change in Q is 10 books. So $MC = 220 \div 10 = \text{Tk } 22$ for each of those books. Fixed cost is Tk 600 in both rows, so it cancels in the subtraction. Marginal cost comes from variable cost alone. A fixed cost, however large, never changes marginal cost.

PAUSE AND CHECK 11.P2 At the workshop, total cost rises from Tk 1,320 at 30 books to Tk 1,560 at 40 books. What is the marginal cost of each of those books?

An owner who sets a price needs the cost of each unit on average. **Average total cost** is total cost divided by the number of units produced. In symbols, $ATC = TC \div Q$. For the tea stall at 25 cups, $ATC = 200 \div 25 = \text{Tk } 8$ a cup. For the workshop at 20 books, $ATC = 1,120 \div 20 = \text{Tk } 56$ a book. Average total cost measures the whole cost, fixed and variable, shared among all the units.

Average total cost splits into two parts. Fixed cost divided by output is the average fixed cost. It falls as output grows, because the same Tk 600 is shared among more books: Tk 60 a book at 10 books, and Tk 10 a book at 60 books. Variable cost divided by output is the average variable cost. At the workshop it is Tk 30, 26, 24, 24, 26 and 29.67 for outputs of 10 to 60 books. Adding the two averages gives ATC. At 10 books, 60 plus 30 gives Tk 90.

Case file, illustrative. For the same workshop, the marginal cost of each book in each ten-book step is Tk 30, 22, 20, 24, 34 and 48. Average total cost at outputs of 10 to 60 books is Tk 90, 56, 44, 39, 38 and 39.67. The table lists these figures beside the average variable cost.

PER-UNIT COSTS OF THE BINDING WORKSHOP, IN TAKA A BOOK

OUTPUT (BOOKS A DAY)	MARGINAL COST OF THE LAST TEN BOOKS	AVERAGE VARIABLE COST	AVERAGE TOTAL COST
10	30	30.00	90.00
20	22	26.00	56.00
30	20	24.00	44.00
40	24	24.00	39.00
50	34	26.00	38.00
60	48	29.67	39.67

Read down each column. Marginal cost falls from 30 to 20 and

Marginal cost asks what one more unit adds to total cost.

GO SLOWLY HERE.

Marginal. Word in use. In daily life, marginal means small or barely enough, as in a marginal gain. In economics, marginal means extra: the change caused by one more unit.

Marginal cost. The extra cost of producing one more unit of output.

At the workshop, marginal cost is each rise in total cost divided by the rise in output.

Average total cost spreads total cost over every unit made.

Average total cost. Total cost divided by the number of units produced.

Average total cost is the sum of two smaller averages, one for each kind of cost.

The second table gives the per-unit costs at each output.

In the table, every cost per

YOU WILL NEED

Chapter 10, 'Production Function and Returns', section 10.1, 'Inputs and Output' (the production function, the short run and the long run); Chapter 10, section 10.2, 'The Law of Diminishing Returns'; and Appendix A1, 'Maths You Will Use'.

Two stalls show the same takings, yet one owner has chosen badly.

The answer depends on counting what the owner gave up, not only what the owner paid.

This chapter sorts costs into kinds, measures each kind and shows how cost changes with output.

A household garden shows that a resource can be costly even when nobody is paid for it.

GO SLOWLY HERE.

The first kind of cost is the one that appears in a receipt or a bank record.

Cost. Word in use. In daily life, cost often means the price a buyer pays. In economics, cost means the seller's expense of making a good, and it includes costs that nobody pays in money.

Explicit cost. A cost that

Two friends open stalls in the same market. At the end of each day, each friend counts Tk 900 left after paying for goods and stall rent. Asha left a job that paid Tk 1,100 a day. Bina had no other offer of work. The two tills show the same figure. Yet one friend has made a poor choice, and the other has not. Which one, and why?

Most students first say that the two stalls are equal, because the tills agree. Look again at Asha. She could have earned Tk 1,100 by working elsewhere, so her stall leaves her Tk 200 worse off each day. Bina had nothing better to do, so for her the same Tk 900 is a pure gain. The puzzle shows that economists use the word *cost* more widely than a shop's account book does. Paying money is one kind of cost. Giving up a better use of your own time or money is another kind.

We do three things in this chapter. First, we separate costs that involve a payment from costs that do not. Next, we study how cost behaves when a firm makes more output in the short run, and we learn to calculate every kind of cost by hand. Last, we see why the cost curves have the shapes they have. Two examples run through the chapter: a small binding workshop, and the shop of Meghna and Sons.

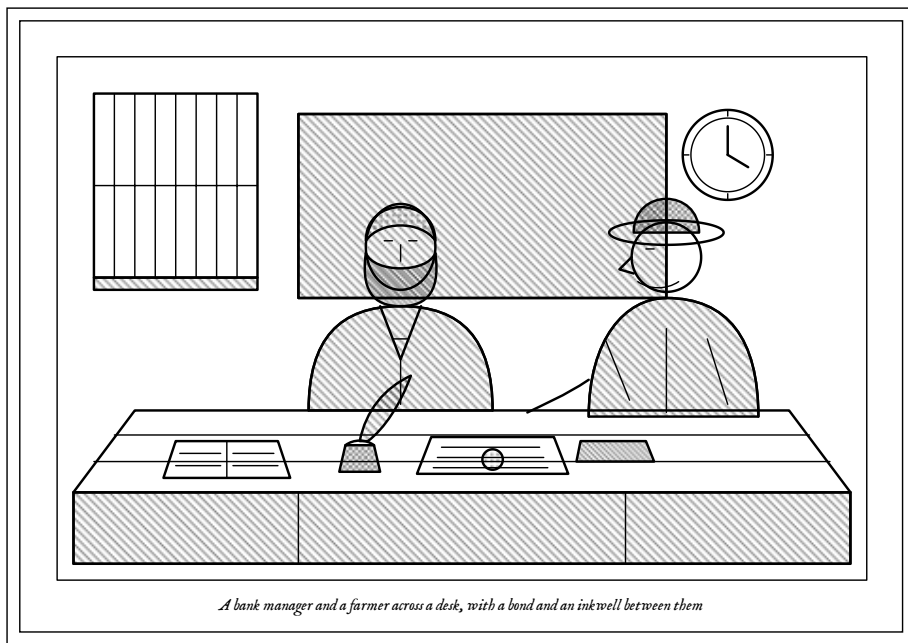
* * *

11.1 EXPLICIT AND IMPLICIT COSTS

Think first of a household, not a firm. A family grows vegetables in its own garden. It pays Tk 300 a month for seeds and water. It pays nothing for the land, which it owns, and nothing for the hours its members work. Is the garden free, apart from the Tk 300? Not quite. The land could be rented to a neighbour, and the hours could earn wages elsewhere. These lost gains are real, even though no money changes hands. Economists call the best alternative given up the *opportunity cost* of a choice.¹

The first kind is easy to see. An **explicit cost** is a cost that involves a payment of money to outsiders. The word *explicit* means openly stated. For example, the rent that a shop pays its landlord each month is an explicit cost, because money leaves the shop and reaches another person. Wages, paper, electricity and interest on a bank loan are also explicit costs. Each one appears in a receipt or a bank record, and an accountant writes every one of them down.

¹ The idea of opportunity cost is taught in Chapter 2, 'Scarcity, Choice and Opportunity Cost'. In this chapter you need only the one-line meaning given in the text.



CHAPTER XVI

Interest and Profit

Why is money paid to wait, and who deserves what is left at the end?



Lenders are paid for letting others use their money, and owners are paid, sometimes, for taking chances. This chapter explains what sets the rate of interest, separates two meanings of profit, and shows why profit arises at all.

15.T2 A new road makes a distant farm easier to reach. What do you expect to happen to the rent of that farm, and why?

WHERE PEOPLE GO WRONG

Many students think that rent raises the price of the goods grown on the land. In Ricardo's theory it is the other way: the marginal land fixes the cost, and rent is the surplus of better land. They often judge a wage by its nominal figure alone, when the real wage shows what the wage buys. They think that economic rent belongs only to land, when any factor can earn it. The fourth error is to expect a firm to hire until the marginal revenue product is zero; it stops when the marginal revenue product falls below the wage.

IN HISTORY

David Ricardo (1772 to 1823) was an English businessman and economist. In 1817 he published *On the Principles of Political Economy and Taxation*, which set out the theory of rent that this chapter describes. At the time, England had high food prices and a fierce argument about the corn laws, which taxed imported grain. Ricardo argued that rent followed from the price of food, and that the poorest land in use paid none.

Wages can also be set by law. In April 2013 the Rana Plaza building, which housed garment factories near Dhaka, collapsed, and more than a thousand people died. After this disaster and a wave of strikes, a government wage board raised the monthly minimum wage for entry-level garment workers from Tk 3,000 to Tk 5,300, effective from December 2013. The minimum wage was revised again in later years. The case shows a wage fixed above the level that the market had set, as discussed in section 15.3.

REVIEW QUESTIONS

- 15.1 Define rent, and explain why the supply of land as a whole is said to be fixed.
- 15.2 A singer is paid Tk 80,000 for an evening and would have accepted Tk 50,000. Find the economic rent.
- 15.3 Three grades of land yield 120, 100 and 80 kg an acre, and the 80 kg land is the marginal land. Find the differential rent of each grade.
- 15.4 A nominal wage of Tk 24,000 a month meets a price index that rises from 100 to 120. Find the real wage.
- 15.5 A nominal wage rises by 5% while prices rise by 8%. Does the real wage rise or fall?
- 15.6 A firm pays Tk 30,000 a month. Successive workers add Tk 45,000, Tk 34,000, Tk 29,000 of revenue. How many should it hire?
- 15.7 Distinguish rent from quasi-rent in one sentence each.

Four common misunderstandings, each corrected.

In 1817 Ricardo explained rent as the gain of better land over the poorest land in use.

In 2013 Bangladesh raised the garment minimum wage after protests and a factory disaster.

YOU WILL NEED

Chapter 9, 'Elasticity of Supply and Market Equilibrium', section 9.2 'Market Equilibrium'; Chapter 11, 'Costs of Production', section 11.2 'Short-Run Costs'; Chapter 11, 'Costs of Production', section 11.3 'The Cost Curves'; and Appendix A1, 'Maths You Will Use'.

Two sellers in one town choose their prices in very different ways.

The difference lies in the surroundings of each seller, not in the seller's character.

The chapter has three steps: sort markets, study one firm, then follow its profit over time.

Picture two sellers in one town. The first sells potatoes from a stall in a long row of stalls, and every stall offers the same potatoes at nearly the same price. The second runs the only pharmacy within ten kilometres. Both sellers wish to earn as much money as possible, yet they behave in opposite ways. The potato seller dares not charge one taka more than her neighbours, because her customers would move to the next stall. The pharmacist may raise a price and still keep many customers.

Why does the same wish lead to such different behaviour? The cause is not the character of the two sellers. It lies in the surroundings of each one: how many rivals there are, whether the goods are the same, and whether a new seller can arrive next month. When the surroundings change, the behaviour changes with them. Economists describe those surroundings in an exact way, so that they can predict what sellers will do.

This chapter takes three steps. First, we sort markets into kinds, so that we know where the potato seller and the pharmacist belong. Second, we study a firm that cannot choose its price and chooses only its output, and we find the output that gives it the largest profit, using the cost data of the binding workshop. Third, we ask what happens to that profit once other firms notice it. Many students find the second step surprising, because they expect every firm to name its own price.

* * *

13.1 KINDS OF MARKET

Suppose you must predict how the sellers in a market will behave, and you may ask only a few questions. Most people begin with the number of sellers. That is a good first question, but not the only one. Economists have found that the answers to a small group of questions, taken together, explain much of what sellers do. A list of those answers describes the market.

The name for that list is **market structure**. A market structure means the set of features of a market, such as the number of sellers and the ease of entry, that shape how firms behave. The row of potato stalls and the village pharmacy have different structures. Once we know the structure, we can say much about the freedom a seller has over the price.

FEATURES THAT DISTINGUISH MARKETS

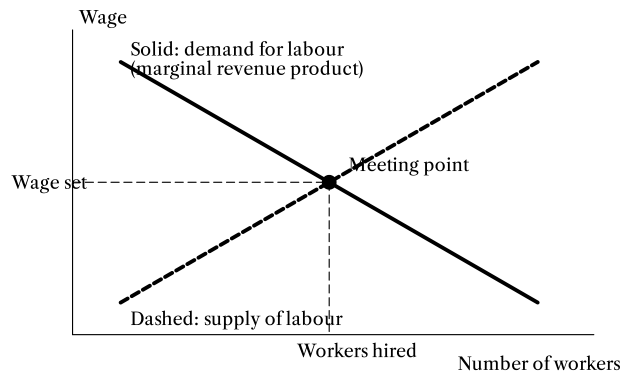
Four features do most of the work. The first is the *number of sellers*: very many, a few, or one. The second is the *kind of good*:

A question: which facts about a market would you ask for, to predict how its sellers behave?

Market structure. The set of features of a market, such as number of sellers and ease of entry, that shape how firms behave.

The list of features that shapes seller behaviour has a name: market structure.

Four features separate one market from another.

FIG. 15.2. *Wage set by the demand for and supply of labour*

Changes follow the pattern of Chapter 9. If firms become more productive or the price of their goods rises, the marginal revenue product rises and demand for labour moves to the right, so the wage and the number employed both rise. If more people enter the occupation, supply moves to the right and the wage falls. A wage fixed by law above the equilibrium wage leaves more workers seeking jobs than firms will fill, though it raises the pay of those who keep their work.

WHY WAGES DIFFER

The same tools explain why wages differ. A surgeon earns more than an assistant for reasons on both sides of the market. On the demand side, a surgeon's work brings in or saves a very large amount, so the marginal revenue product is high. On the supply side, few people can qualify, since the training is long, costly and demanding, so the supply is small. A high demand meeting a small supply gives a high wage.

Other causes follow the same logic. Dangerous or unpleasant jobs pay more, because few people accept them at an ordinary wage. Workers with scarce skills are in short supply. Trade unions raise wages by limiting supply or by bargaining. A firm that is the only large employer in a town, a position like the monopoly of Chapter 14, may pay less than a competitive market would, because workers have nowhere else to go. Differences in wages are therefore not random; each can be traced to demand, supply or both.

Wages are sometimes fixed by law. A minimum wage set above the equilibrium wage raises the pay of workers who keep their jobs, but at that higher wage firms demand fewer workers than seek work. Whether jobs are lost, and how many, depends on how strongly firms respond to the higher wage; the wider the gap between the legal wage and the equilibrium wage, the larger the likely loss. Governments weigh this against the gain to the workers who are paid more.

Before moving on. A firm hires while the marginal revenue

Recall: changes in equilibrium, Chapter 9, section 9.3 'Changes in Equilibrium'.

Wages differ because the demand and supply differ between occupations.

Other causes work through the same two sides.

Recall: monopoly, Chapter 14, section 14.1 'Monopoly'.

A minimum wage fixed above the equilibrium wage helps those hired and may leave others without work.

Before moving on: wages follow the marginal revenue

Each of the four conditions removes one kind of power from the seller.

Price taker. A firm that must accept the market price because its own sales are too small to affect it.

Because of these conditions, each firm must accept the price that the market gives.

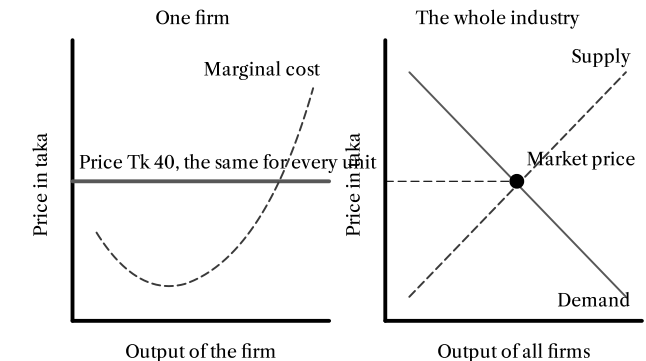
How to read the first chart: two panels share a price scale but not a quantity scale.

therefore a *model*: a simple picture that shows the main forces in their cleanest form.¹

Each condition removes one kind of power from the seller. With very many sellers, each firm supplies a tiny part of the total. With identical goods, a buyer has no reason to prefer one seller. With free entry, new firms may start at any time. With full information, every buyer knows the price in the next stall and will not pay more.

The result is a firm of a special kind: a **price taker**. A price taker means a firm that must accept the market price because its own sales are too small to affect it. Suppose the market price of a notebook is Tk 40. A workshop that makes notebooks can sell any quantity it is able to make at Tk 40, and none at Tk 41. It has no reason to sell below Tk 40, since all its output finds a buyer at Tk 40. For this firm the price is a fact, like the weather, and it cannot be changed.

The first chart shows a firm and its industry side by side. In each panel the vertical axis shows the price in taka, and the horizontal axis shows quantity. A point on a line shows one price and the quantity that goes with it. The two panels use the same price scale, so the same height means the same price. Their quantity scales differ, because one firm makes only a tiny part of the industry's total. On the right, the industry faces a falling demand line and a rising supply line, which cross at the market price. On the left, the single firm faces a flat line at that price.

FIG. 13.1. *Firm and industry side by side: the market sets the price, and the firm accepts it*

PAUSE AND CHECK 13.P1 The market price is Tk 40. A firm in perfect competition asks Tk 42 for its good. What happens to its sales, and why?

MARGINAL COST EQUALS PRICE

Because the market fixes the price, the firm has one decision left:

¹ A model leaves out detail on purpose. Its value lies in the forces it shows, and not in an exact match with any one market.

GO SLOWLY HERE.

A price taker has one decision left: how much to make

* * *

15.2 WAGES

THE WAGE

Labour is the work of people, and it is paid. A **wage** means the payment to labour for its work over a period. It may be stated by the hour, the day or the month. A shop assistant paid Tk 35,000 a month earns a wage, and so does a surgeon paid Tk 400,000 a month. In daily speech a salary and a wage are different; in economics the two are the same kind of payment.

NOMINAL WAGE AND REAL WAGE

A wage is first stated in money, but money is only useful for what it buys. If the prices of goods double, the same money buys half as much. Economists therefore separate the money amount from its purchasing power.

The **nominal wage** is a wage measured in money, without allowing for changes in prices. Tk 35,000 a month is a nominal wage. It is the figure on the pay slip.

The **real wage** is a wage measured by the goods and services it can buy. To find it, divide the nominal wage by a price index, which is a number showing the level of prices compared with a base year set at 100. In words: real wage = nominal wage ÷ (price index ÷ 100). In symbols, real wage = $W \div (P \div 100)$, where W stands for the nominal wage and P for the price index.

Case file, illustrative. A shop assistant at Meghna and Sons earns a nominal wage of Tk 35,000 a month. The price index rises from 100 to 112. The real wage is $35,000 \div 1.12 = \text{Tk } 31,250$.

Given: a nominal wage of Tk 35,000 a month, and a price index that rises from 100 to 112. *Find:* the real wage, in the prices of the base year. *Formula:* real wage = nominal wage ÷ (price index ÷ 100). *Steps.* The index divided by 100 is $112 \div 100 = 1.12$. Then $35,000 \div 1.12 = 31,250$. *Answer:* the real wage is Tk 31,250 a month, in base-year prices. *Check:* prices have risen 12%, so the wage should buy less than Tk 35,000 did before, and Tk 31,250 is lower.

The result carries a lesson. Suppose the assistant is given a rise of 10%, to Tk 38,500, while the price index rises to 112. The real wage is $38,500 \div 1.12 = \text{Tk } 34,375$ in base-year prices, which is below the first figure of Tk 35,000 in the same prices. The assistant has more money and less purchasing power. A common slip is to judge a wage by the nominal figure alone.

The word wage is used loosely in daily life.

Wage. Word in use. In daily life, the weekly pay of a manual worker, as distinct from a salary. In economics, any payment to labour, whatever its name.

Wage. The payment to labour for its work over a period.

A wage is the payment to labour for its work over a period.

A wage in money can hide what the worker can actually buy.

Nominal wage. A wage measured in money, without allowing for changes in prices.

A nominal wage is the money amount, unadjusted for prices.

Real wage. A wage measured by the goods and services it can buy.

A real wage measures the goods and services the wage buys.

The next record sets out the figures.

The Steps follow the usual order, from the given figures to a check.

The nominal wage can rise while the real wage falls.

The rule has limits: marginal cost must be rising, and the data come in steps.

Before moving on, recall the price taker and the output rule.

A puzzle: the workshop earns a profit, so would that profit last?

Equilibrium of the firm.

The output at which a firm earns the largest possible profit and has no wish to change.

The best output of a firm has a name: the equilibrium of the firm.

The case file states the workshop's revenue, cost and profit at its best output.

A test: at 40 units and at 60 units the profit is lower, so the workshop wishes no change.

Profit. Word in use. In daily life, any money gained, or the money left after the bills are paid. In economics, revenue minus all costs, including the value of the owner's own time and money; Chapter 16, 'Interest and Profit', section 16.2 'Profit' explains it fully.

In this chapter, profit means total revenue minus total cost.

In the short run, the price decides whether the firm gains

The rule has two limits. First, the marginal cost must be *rising* where it meets the price. A marginal cost that meets the price while it is still falling does not mark the best output, because the next units cost less and add more profit. The workshop's costs fall over the first three steps and rise afterwards, so only the rising part can give the answer. Second, the data come in steps of 10 units. A smooth curve would cross the price line between 50 and 60 units, but the workshop sells whole steps, and the best full step is 50.

Before moving on. Perfect competition is a market with very many buyers and sellers, identical goods, free entry and full information. A price taker accepts the market price and chooses only its output. The firm raises output while marginal cost is below the price and stops where marginal cost equals the price.

* * *

13.3 EQUILIBRIUM OF FIRM AND INDUSTRY

At a price of Tk 40 the binding workshop earns a profit. Other firms can see this, and in perfect competition they are free to enter. Will the profit last for ever? To answer, we first need to be exact about the workshop's best output, and about what the word profit means in this chapter.

The **equilibrium of the firm** means the output at which a firm earns the largest possible profit and has no wish to change. The idea is the same as in Chapter 9, where equilibrium meant a balance of forces, but now it describes one firm. At the best output, a larger output gives less profit and a smaller output gives less profit, so the firm stays where it is.

Case file, illustrative. At the market price of Tk 40, the binding workshop produces 50 units. Total revenue is $50 \times 40 = 2,000$. Total cost is 1,900. Profit is $2,000 - 1,900 = \text{Tk } 100$.

We can test the claim. At 40 units, total revenue is $40 \times 40 = 1,600$. Total cost is the cost at 50 units less the cost of the fifth step: $1,900 - 340 = 1,560$. Profit is $1,600 - 1,560 = \text{Tk } 40$. At 60 units, total revenue is $60 \times 40 = 2,400$, and total cost is $1,900 + 480 = 2,380$, so profit is Tk 20. Both figures are below Tk 100. The workshop has no wish to change, so 50 units is its equilibrium.

In this chapter, *profit* means total revenue minus total cost. Total revenue is the price times the quantity sold. If total revenue is larger than total cost, the firm earns a profit. If it is smaller, the firm makes a loss. The word has an everyday meaning too, and the two meanings differ in what they count as a cost. The note above gives both.

SHORT-RUN PROFIT AND LOSS

The short run is a period too short for new firms to enter or for old firms to leave. During it the market price decides the result for

This observation leads to a wider idea. **Economic rent** means the payment to any factor above the least needed to keep it in use. A factor is any resource used in production, such as land, labour or a machine. Suppose a famous singer is paid Tk 100,000 for an evening and would have agreed to sing for Tk 40,000. The least needed is Tk 40,000, and the economic rent is Tk 60,000. For land as a whole the least needed is nothing, since it exists whatever is paid, so the whole payment is economic rent.

For one plot, the position differs. A plot of farmland has another use, and its owner could sell it for building. The least needed to keep it in farming is what that other use would pay, which Chapter 11 would call an implicit cost. Only the part above that is economic rent. This is a common source of confusion, and it is why economists speak of economic rent for any factor and not only land.

Viewpoints differ. For the owner, rent is income from a fixed asset that needs no extra effort to supply. For the tenant, rent is a payment that must be covered by the sales of the shop or the crop. For the buyer of the goods, rent is not a reason for a high price, since the site's advantage creates the rent and not the other way round. A tenant who pays a high rent for an excellent site is paying for its advantage, and could not make the same profit from a poor site at a low rent.

Before moving on. Rent is the payment to the owner of land for its use. Land as a whole is fixed in supply, so its whole payment is economic rent. Economic rent is the part of any payment above the least needed to keep a factor in use.

RICARDIAN THEORY OF RENT

Why does one plot pay more rent than another? The English economist David Ricardo gave the classic answer in 1817. His theory rests on a simple picture: land differs in quality, and the best land is used first.

Imagine a district with three grades of land. A farmer using one acre of Grade A land harvests 3,000 kg of rice. On Grade B the same work and seed yield 2,500 kg, and on Grade C they yield 2,000 kg. Suppose that Grade C is the poorest land still worth farming, so it pays no rent: it is the marginal land. A farmer will not pay more for Grade B than the extra 500 kg it gives over Grade C, because Grade C is available for nothing. For the same reason Grade A will rent for the extra 1,000 kg. Rent equals the difference in yield between a plot and the marginal land.

To read the first figure, look at the stacked bars. Each bar is one grade of land, and its height is the yield of one acre. The lower, hatched part of each bar is the yield of the marginal land, which every grade produces. The dotted part above it is the extra yield, and that is the rent. Grade C has no dotted part, because it is the marginal land and pays no rent.

Economic rent. The payment to any factor above the least needed to keep it in use.

Economic rent is the part of a payment above the least needed to keep a factor in use.

Recall: explicit and implicit costs, Chapter 11, section 11.1 'Explicit and Implicit Costs'.

The owner, the tenant and the buyer of the goods each see rent differently.

Before moving on: rent is a payment for land; economic rent is a surplus above the least needed.

Differences in the quality of land explain why some plots pay more.

Rent arises because better land produces more at the same cost.

The figure shows rent as the part of the yield above the marginal land's yield.

step equals 10 times its marginal cost. *Steps.* (1) Extra revenue of each step: $10 \times 30 = 300$. (2) Extra cost of the steps: 300, 220, 200, 240, 340, 480. (3) Extra profit of the steps: 0, +80, +100, +60, -40, -180. (4) The fifth step reduces profit, so the best output is 40 units. (5) Total revenue: $40 \times 30 = 1,200$. (6) Total cost at 40 units: $1,900 - 340 = 1,560$. (7) Profit: $1,200 - 1,560 = -360$. *Answer.* The workshop makes 40 units and a loss of Tk 360. *Check.* At 30 units the loss would be $(1,560 - 240) - 900 = 420$, and at 50 units it would be $1,900 - 1,500 = 400$. Both are larger than 360, so 40 units gives the smallest loss.

The second example joins this chapter to the cost ideas of Chapter 11. It supposes that other firms have the same costs as the workshop, and that entry is free.

Given. Total cost is 1,560 at 40 units, 1,900 at 50 units and 2,380 at 60 units. The market price is Tk 40, and the workshop earns Tk 100 above normal. *Find.* The price at which the profit above normal disappears. *Formula.* In words: cost per unit equals total cost divided by quantity. In the long run, the price equals the lowest cost per unit. *Steps.* (1) At 40 units: $1,560 \div 40 = 39$. (2) At 50 units: $1,900 \div 50 = 38$. (3) At 60 units: $2,380 \div 60 = 39.67$, to two decimal places. (4) The lowest cost per unit is Tk 38, at 50 units. (5) At Tk 38 the best output is still 50 units, since the marginal cost of the fifth step (34) is below 38 and that of the sixth (48) is above it. (6) Total revenue: $50 \times 38 = 1,900$, which equals total cost. *Answer.* The long-run price is Tk 38. The workshop makes 50 units and earns normal profit only. *Check.* A fall of Tk 2 in the price takes $2 \times 50 = \text{Tk } 100$ from the revenue. That equals the profit above normal at Tk 40, so no profit above normal remains.

Three slips cause most wrong answers. First, forgetting that the extra cost of a 10-unit step is 10 times its marginal cost. Second, stopping at the first output where the price equals the marginal cost, when the marginal cost is still falling. Third, comparing the price with total cost, when it should be compared with marginal cost.

* * *

SUMMARY

A market structure is the set of features of a market that shape how firms behave. Perfect competition has very many buyers and sellers, identical goods, free entry and full information. In it, each firm is a price taker, which accepts the market price and chooses only its output. The firm raises output until marginal cost equals the price, and that output is the equilibrium of the firm. In the short run the firm may earn a profit above normal or make a loss. In the long run, free entry and exit leave every firm with normal profit, at a price equal to the lowest cost per unit.

THINK IT THROUGH

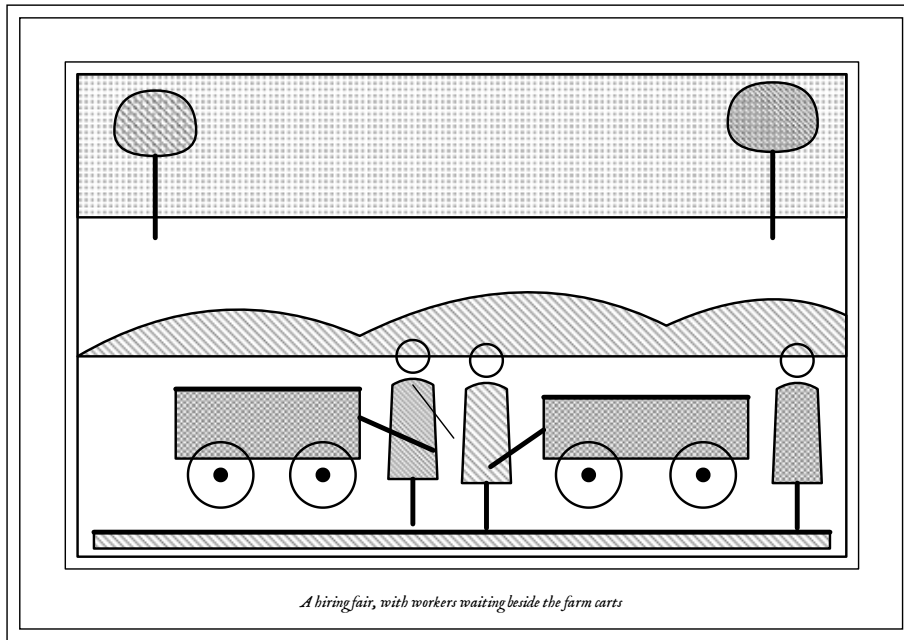
Recall: cost curves, Chapter 11, section 11.3 'The Cost Curves'.

Worked example 2 finds the price at which the profit above normal disappears.

The steps of worked example 2, from the data to the check.

Common slips: step costs, falling marginal cost and mixing totals with unit figures.

The chapter in a few lines.



CHAPTER XV

Rent and Wages

Why is one field paid for, and one worker paid more than another?



This chapter explains why land earns a payment that no one needs to be tempted to give, how the payment to labour is measured in money and in goods, and how demand and supply set a wage and explain why wages differ.

13.4 In the notebook market, the quantity demanded is 80 minus the price and the quantity supplied equals the price. Demand then shifts to 100 minus the price. Find the new market price, then find the workshop's best output and profit at that price, and say what you expect to happen in the long run.

13.5 A town has thirty tailors. Each offers clothes in a different style, and anyone may open a new shop. Using the table in section 13.1, name the kind of market and give the features that support your answer.

THINK IT THROUGH

14.T1 Is a lower price always better for consumers? Give one reason from the chapter why the answer may be no.

14.T2 Why will a seller with positive marginal cost never choose an output at which marginal revenue is negative? Use the 50-unit row of the table to test your answer.

WHERE PEOPLE GO WRONG

Some students think a monopolist can charge any price. It cannot. It controls its price only up to the limit set by the demand curve: a higher price brings fewer buyers. The seller in the case file chooses Tk 50, not Tk 500.

Some students use the price as the marginal revenue of a monopolist. For a price taker the two are equal, but for a price maker marginal revenue is smaller, because a cut in price applies to every unit sold.

Some students treat monopolistic competition as a weak form of monopoly. It differs in kind. It has many sellers and free entry, so extra profit does not last, while a monopoly is protected by barriers.

Some students think firms in an oligopoly must agree with one another. The defining feature is interdependence, not agreement. Firms may compete hard, and agreements are often unlawful and fragile.

IN HISTORY

At Kimberley, in what is now South Africa, many separate owners worked diamond claims in the 1870s and 1880s. Cecil Rhodes, a British businessman who had come to South Africa as a young man, set out to buy the claims and join them under one owner. His chief rival was Barney Barnato, another Briton, whose firm owned much of the Kimberley mine.

In March 1888 the rival holdings were merged to form De Beers Consolidated Mines, with Rhodes as its first chairman. In July 1889 the firm paid £5,338,650, then an unprecedented sum, for the Kimberley mine, and in the same month it won control of two more mines, Bultfontein and Dutoitspan. It then controlled most of the world's supply of diamonds, the first barrier of section 14.1. That barrier did not hold for ever: later, new mines in other places came into production and loosened the firm's hold.

REVIEW QUESTIONS

14.1 Total revenue is Tk 1,500 at 30 units and Tk 1,600 at 40 units. Find the marginal revenue per unit for the step from 30 to 40 units.

14.2 A seller faces price = 60 – quantity and has a fixed cost of Tk 100. Total revenue is 500, 800, 900 and 800 at 10, 20, 30 and 40

A monopolist cannot charge any price it likes.

Marginal revenue is not the same as price for a price maker.

Monopolistic competition is not a mild monopoly.

Oligopoly does not mean that firms agree.

A British businessman joined many diamond owners into one firm.

In 1888 the rival holdings merged into De Beers Consolidated Mines.

YOU WILL NEED

Chapter 13 'Market Structure and Perfect Competition', section 13.1 'Kinds of Market'; Chapter 13 'Market Structure and Perfect Competition', section 13.2 'The Perfectly Competitive Firm'; Chapter 11 'Costs of Production', section 11.2 'Short-Run Costs'; Appendix A1 'Maths You Will Use'.

One seller seems able to charge any price, yet something stops it.

The seller must weigh the price she asks against the quantity she sells.

Most real markets lie between the two extremes.

A town has one shop that sells a rare title which every student of a certain subject must read. No other shop stocks it. The owner thinks: "I am the only seller, so I may charge Tk 1,000 a copy." Why does she not do so? At Tk 1,000 almost no student would buy, and her shelf would stay full. At a very low price many students would buy, but she would earn little on each copy. Somewhere between the two lies a price that brings her the most profit. How does she find it?

Chapter 13 described the opposite case. A firm in perfect competition is a price taker: it accepts the market price and chooses only its output. A single seller faces a harder and more interesting choice. Every time she lowers her price to sell more, she earns less on the copies that she could already sell. So she must weigh price against quantity, and the weighing needs a new idea, which we meet in section 14.1.

Few markets look like either extreme. A street of tea stalls has many sellers, yet each stall is a little different. Mobile telephone service in many countries is supplied by a handful of large firms. This chapter therefore studies four markets in turn: the single seller, the many sellers whose goods differ, the few large sellers, and then all four side by side. Many students find the long list of market names confusing. We give each name one plain definition and one picture, and we compare the markets only at the end.

* * *

14.1 MONOPOLY

Begin with a village on a wide river. One ferry carries people across, and no bridge or other boat exists. The ferry owner does not fear that a neighbouring ferry will take her passengers, because there is none. She can choose her fare, and the villagers must pay it or stay on their side. Her position is the heart of this section: a seller with the whole market to itself.

A **monopoly** is a market in which one seller supplies a good with no close substitute. The ferry across the river is a monopoly, and so is the only pharmacy on a remote island. The word *close* matters: a good has a close substitute if consumers would readily switch to it when the price rises. Swimming is not a real choice for the villagers, so the ferry has none. The size of the market does not matter; a monopoly may serve a village or a nation.

Monopoly. A market in which one seller supplies a good with no close substitute.

Monopoly. Word in use. In daily life, people call any very large firm a monopoly, and a board game carries the same name. In economics, it means one seller with no close substitute, however small the market.

One seller with no close

14.4 COMPARING MARKETS

The table below sets the four markets side by side. Read one column at a time to see what defines each market, and read one row at a time to see how a single feature changes as we move from many sellers to one. The figure that follows shows the number of sellers as dots, and the shapes show whether the goods are alike or different.

THE FOUR MARKET STRUCTURES COMPARED

FEATURE	PERFECT COMPETITION	MONOPOLISTIC COMPETITION	OLIGOPOLY	MONOPOLY
Sellers	Very many	Many	Few	One
Goods	Identical	Similar but different	Identical or different	No close substitute
Entry	Free	Free	Hard	Blocked
Price	Price taker	Price maker, small power	Price maker, watches rivals	Price maker, large power

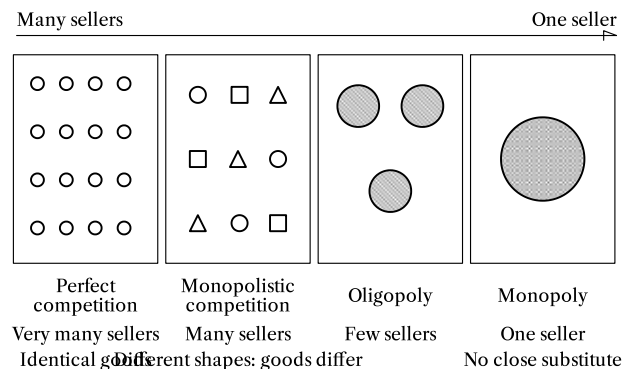


FIG. 14.2. Four market structures: very many, many, few and one seller

Suppose, as an illustration, that a competitive market faced the same demand and the same marginal cost as the seller of the case file. Competitive sellers push output to the point where price equals marginal cost. At 40 units the price is Tk 40 and the marginal cost of the step is Tk 24, so output rises. At 50 units the price is Tk 30 and the marginal cost is Tk 34, so output stops at 40. The competitive market would sell 40 units at Tk 40, while the monopolist sells 30 units at Tk 50: ten fewer units, at a price Tk 10 higher.

Is a lower price always better? Not always. A competitive market tends to give the lowest price for a given good, but one large firm may serve everyone at a lower cost for each unit than many small firms could. Monopolistic competition gives consumers variety that a market of identical goods cannot. Price is one measure of how well a market serves consumers; quantity, quality, variety and the

A table compares the four markets on four features.

THE SELLER'S PRICE, REVENUE AND COSTS (TAKA)

OUTPUT	PRICE	TOTAL REVENUE	MARGINAL REVENUE	MARGINAL COST
10	70	700	70	30
20	60	1,200	50	22
30	50	1,500	30	20
40	40	1,600	10	24
50	30	1,500	-10	34

The best output is where marginal revenue meets marginal cost.

The figure draws the same facts as three lines.

How much should she sell? Compare, step by step, what a step adds to revenue with what it adds to cost. A step is worth taking while marginal revenue is at least marginal cost, because it then adds at least as much to revenue as to cost. Look down the table. The step to 30 units adds Tk 30 a unit to revenue and Tk 20 to cost, so it adds profit. The step to 40 units adds only Tk 10 to revenue but Tk 24 to cost, so it loses profit. The best output is 30 units, and the price comes from the demand curve: Tk 50. This is the rule that Chapter 13 used for the competitive firm, with marginal revenue in place of price.²

The figure shows the same facts. The horizontal axis counts output in units, and the vertical axis counts taka. The falling straight line is the demand curve: each point gives the price at which that output sells. The steeper line below it is marginal revenue, which falls twice as fast and reaches zero at 40 units. The curved line is marginal cost. Where marginal revenue meets marginal cost, at about 30 units, we climb to the demand curve to read the price, Tk 50. The shaded rectangle is the profit: the gap between the price of Tk 50 and the average cost of Tk 44 ($1,320 \div 30$) is Tk 6, and Tk 6 times 30 units is Tk 180.

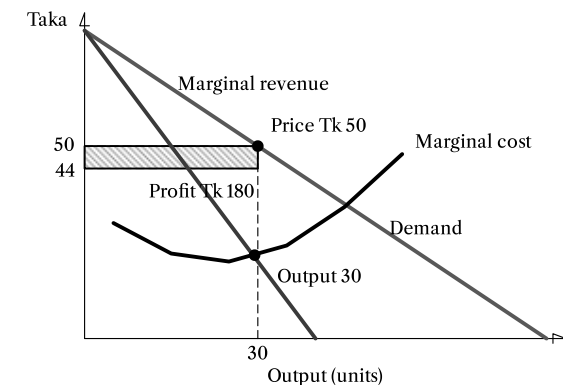


FIG. 14.1. The monopolist's demand, marginal revenue and marginal cost, with the best output and its profit

The theory has exceptions. First, a monopolist may lose money if

² Recall: marginal cost, Chapter 11, 'Costs of Production', section 11.2 'Short-Run Costs'; the output of largest profit is taught in Chapter 13, 'Market Structure and Perfect Competition', section 13.3 'Equilibrium of Firm and Industry'.

In the case file, the monopoly sells less at a higher price than a competitive market would.

Lower price is one measure of a market, not the only one.

A monopoly is not sure of

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YOU WILL NEED

Chapter 15, 'Rent and Wages', section 15.1, 'Rent'; Chapter 11, 'Costs of Production', section 11.1, 'Explicit and Implicit Costs'; Chapter 13, 'Market Structure and Perfect Competition', section 13.3, 'Equilibrium of Firm and Industry'; and Appendix A1, 'Maths You Will Use'.

A neighbour with spare money must decide whether to lend it for nothing.

A second puzzle asks whether the money left after repayment is truly a gain.

This chapter studies the price of money and the reward of the owner.

A friendly loan shows why a lender wants more back than was lent.

GO SLOWLY HERE.

Interest is the price paid for the use of money over time.

Interest. Word in use. In daily life, interest often means curiosity or attention, as in an interest in music. In economics, it means a payment for the use of money.

Interest. The payment a borrower makes to a lender for the use of money over time.

The bank's loan to the shop is

In spring, a farmer needs Tk 100,000 to buy seed and tools. His neighbour has exactly that sum saved. She has three choices. She can keep the money in a box, give it to him for nothing, or hand it over and ask for more money back after the harvest. Most neighbours choose the third. Why do they ask for more, and how much more is fair?

A second puzzle follows the farmer to the autumn. He sells the harvest, repays the loan with the extra amount, and has Tk 20,000 left. Is that Tk 20,000 a gain? Not necessarily. If he could have earned more by working for another farmer, the Tk 20,000 may be less than he gave up. Both puzzles ask the same thing: what is a fair payment for money, and what is a fair reward for running a business?

We study two payments in this chapter. The first is the payment for the use of money, which is called interest. The second is the reward of the person who runs the firm and takes the chances, which is called profit. We use the shop of Meghna and Sons and a famous lender of Bangladesh as our examples.

* * *

16.1 INTEREST

Begin with a loan between friends. You lend a friend Tk 1,000 today, and he promises to repay in one year. During that year you cannot spend the money. You also run a small risk that he will not repay. Most lenders will not accept Tk 1,000 back, because they would be worse off than if they had kept the money. They ask for a little more, perhaps Tk 1,100. The extra Tk 100 is the lender's reward for waiting and for taking the risk.

The extra Tk 100 has a name. **Interest** is the payment a borrower makes to a lender for the use of money over time. The payment is stated as a rate, which is the interest for one year divided by the sum borrowed, written as a per cent. In the friend's loan, Tk 100 on Tk 1,000 is a rate of 10 per cent a year. Interest is like the rent of land, which is a payment for the use of something owned by another person.¹

Case file. illustrative. Meghna and Sons holds a bank loan of Tk 1 Rent as a payment for the use of land is taught in Chapter 15, 'Rent and Wages', section 15.1 'Rent'. Interest is the matching payment for the use of money.

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Interest looks different to the lender, the borrower and the owner of a firm.

The section closes with a recap and a first test.

A harvest sale shows that the money left over can be counted in two ways.

GO SLOWLY HERE.

The accountant's measure subtracts only the costs that were paid in money.

Profit. Word in use. In daily life, profit means any gain or money left in the till. In economics, it has two exact meanings, which differ by the owner's implicit costs.

Accounting profit. Revenue minus explicit costs only.

The economist's measure also subtracts what the owner's own resources could have earned.

Economic profit. Revenue minus all costs, explicit and implicit, including normal profit.

Normal profit is the least profit that keeps an owner in the

avoid a fixed rate of interest and share the results of the venture with the borrower instead.

The viewpoint changes the meaning. To the lender, interest is income for waiting. To a household that borrows, interest is a cost of buying now instead of later. To the manager of a firm, interest on a loan is an explicit cost, and the manager compares it with the extra output that the borrowed money can buy. A loan is worth taking only when the money earns more than the interest it costs.

Before moving on. Interest is the payment for the use of money, stated as a yearly rate. The loanable funds theory sets the rate where savers' offers equal borrowers' demands. The liquidity preference theory says interest rewards people for giving up cash. Rates differ with risk, length and size. You should now be able to say why a lender asks for more than was lent.

PAUSE AND CHECK 16.P1 A bank lends Tk 200,000 for one year at 9 per cent a year. How much interest does the borrower pay, and how much does the borrower repay in total?

* * *

16.2 PROFIT

Return to the farmer. He sells his harvest for Tk 300,000. This money received from sales is called revenue. He pays Tk 250,000 for seed, labour, fuel and the interest on his loan. He has Tk 50,000 left. A bookkeeper would call that his gain. But the farmer also worked for months on his own land, and his capital was tied up in the farm. The Tk 50,000 does not yet say whether the year was worth his effort. We need two measures, not one.

The accountant's measure is the simpler one. **Accounting profit** is revenue minus explicit costs only. An explicit cost is a cost that involves a payment of money to outsiders, such as wages, rent and interest.² The farmer's accounting profit is Tk 300,000 minus Tk 250,000, which is Tk 50,000. This is the profit that appears in a firm's accounts and on which taxes are usually based. Rules for tax differ by country, so a student should always name the country whose rules apply.

The economist's measure goes further. **Economic profit** is revenue minus all costs, explicit and implicit, including normal profit. An implicit cost is the value of resources the owner supplies, such as the owner's time and money. If the farmer could have earned Tk 60,000 as a hired worker, that sum is an implicit cost. His economic profit is Tk 50,000 minus Tk 60,000, which is a loss of Tk 10,000. The same year looks like a success in the accounts and a poor choice in the economist's reckoning.

The words *normal profit* need a reminder. Normal profit is the least profit a firm must earn to stay in business, covering the ²Explicit and implicit costs are taught in Chapter 11, 'Costs of Production', section 11.1 'Explicit and Implicit Costs'.

³Normal profit is taught in Chapter 13, 'Market Structure and Perfect Competition', section 13.3 'Equilibrium of Firm and Industry'.

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 William Stanley Jevons, The Theory of Political Economy, 1871

Profit rewards the owner for taking on the chance of loss.

Risk bearing. The act of taking on the chance of loss in business, which profit rewards.

A second view treats profit as a payment for the owner's own contribution.

A third view says that new methods create profit until rivals copy them.

A fourth view says that power over price can protect profit.

The table pairs each view with the case it explains best and the case it misses.

Supernormal profit tends to attract rivals unless something blocks them.

The four views disagree only because they describe different firms.

The first view is the oldest. **Risk bearing** is the act of taking on the chance of loss in business, which profit rewards. A firm that makes goods before knowing whether anyone will buy them may lose its money. Frank Knight, an American economist, argued in 1921 that some chances cannot be known in advance or insured, and that profit is the reward for facing them. Limit: the view explains why a risky venture should earn more on average, not why every owner who takes a risk is paid.

The second view is called the marginal productivity view. It treats the owner, who organises the land, labour and capital, as one more input. The owner's pay is the extra output that the owner's organising adds, as with any other input. Limit: the contribution of one owner is hard to measure, so the view is hard to test.

The third view comes from Joseph Schumpeter, an Austrian economist who wrote in 1911. He argued that profit is the reward of innovation, which means introducing a new good or a new method. The innovator's costs are lower, or the good is new, so economic profit appears. Rivals then copy the idea, and the profit shrinks to zero. Limit: many firms earn profit without any new idea.

The fourth view is monopoly. A firm that is protected from rivals, because entry is difficult, can keep its price above its costs for a long time. Its supernormal profit lasts, because no new firm can enter to share it. Limit: this explains profit that is protected, but not profit in markets with many rivals.

FOUR VIEWS OF WHY PROFIT OCCURS

VIEW	SOURCE OF PROFIT	MAIN LIMIT
Risk bearing	Reward for facing chance of loss	Does not pay every risk-taker
Marginal productivity	Pay for the owner's own contribution	Hard to measure for one owner
Innovation	New methods and goods, until copied	Many profits need no new idea
Monopoly	Protection from rivals	Does not explain profit among many rivals

To read a row, take the view in the first column and ask what source it names in the second. Then check the third column for its weakness. No row is complete alone. A careful economist uses all four views, and asks which fits the firm being studied.

Two of the views end in the same place. Supernormal profit signals that an activity pays more than its alternatives, and it invites new firms to enter. As they enter, the extra supply lowers the price, and the supernormal profit shrinks. Only protection from rivals, such as a new idea that others cannot copy or a market that new firms cannot enter, can keep it for long. This is why innovation and monopoly are linked with lasting profit.

The views are better understood as answers to different questions. A farmer who plants before the rains faces risk. A manager of a large firm supplies organising skill. A new software firm relies on innovation. A single seller of a protected good relies on monopoly.

CHAPTER XV · RENT AND WAGES

15.p1 The 60-bag land pays no rent. The 70-bag land pays $70 - 60 = 10$ bags, and the 90-bag land pays $90 - 60 = 30$ bags.

15.t1 Demand: a surgeon's work brings in or saves a great deal, so the marginal revenue product is high. Supply: few people can qualify because training is long and costly. High demand and small supply give a high wage.

15.t2 The farm's yield net of transport falls less, or its sale price rises, so its advantage over the marginal land grows and its rent rises. Demand for its use rises while its supply is fixed.

15.1 Rent is the payment to the owner of land for its use. Nobody can make more land, so the total supply does not respond to payment.

15.2 Economic rent = $80,000 - 50,000 = \text{Tk } 30,000$.

15.3 Marginal land: 0. The 100 kg grade: $100 - 80 = 20$ kg. The 120 kg grade: $120 - 80 = 40$ kg.

15.4 $120 \div 100 = 1.2$, and $24,000 \div 1.2 = \text{Tk } 20,000$.

15.5 The real wage falls. Money rises by 5% but prices rise by 8%, so purchasing power is lower.

15.6 Hire two. The first two add 45,000 and 34,000, above the wage of 30,000. The third adds 29,000, which is below it.

15.7 Rent is the lasting payment to the owner of land, whose supply is fixed. Quasi-rent is a temporary surplus of a man-made asset while its supply is fixed in the short run, and it disappears as new assets are built.

CHAPTER XVI · INTEREST AND PROFIT

16.p1 Interest = $200,000 \times 0.09 = \text{Tk } 18,000$. Total repayment = $200,000 + 18,000 = \text{Tk } 218,000$.

16.p2 Accounting profit = $900,000 - 700,000 = \text{Tk } 200,000$. Economic profit = $200,000 - 150,000 = \text{Tk } 50,000$, which is super-normal profit.

16.t1 The risky venture may end in loss, and owners will not accept that chance unless the average reward is larger. If it earned no more, few people would take the risk. Profit is the reward for facing the chance of loss.

16.t2 The small trader is a riskier borrower, because the chance of non-repayment is higher. A small loan also costs almost as much to administer as a large one, so the cost per taka lent is higher.

16.1 Interest = $50,000 \times 0.10 = \text{Tk } 5,000$. Total repayment = $50,000 + 5,000 = \text{Tk } 55,000$.

16.2 9.5 per cent is 0.095. Interest = $450,000 \times 0.095 = \text{Tk } 42,750$.

16.3 Interest for a year is $600,000 \times 0.11 = \text{Tk } 66,000$. Three months is $3 \div 12$ of a year, so interest = $66,000 \times 3 \div 12 = \text{Tk } 16,500$.

16.4 Forgone interest = $0.06 \times 2,000,000 = \text{Tk } 120,000$. Implicit costs = $300,000 + 120,000 = \text{Tk } 420,000$. Economic profit = $420,000 - 420,000 = \text{Tk } 0$. This is normal profit only.

explained by risk bearing, the owner's contribution, innovation and monopoly power.

* * *

THINK IT THROUGH

16.T1 Two ventures have the same expected sales and costs, but one is far more risky. Using the risk bearing view, explain why the risky venture should be expected to earn more on average.

16.T2 A bank charges 8 per cent a year to a large firm and 18 per cent a year to a small trader. Give two reasons for the difference.

* * *

WHERE PEOPLE GO WRONG

Profit in the accounts is not the same as economic profit.

Interest is a price, not a punishment.

Normal profit is a cost, not a gain.

Each view of profit fits some firms only.

Mistake: "A profit of Tk 750,000 means the shop is doing well." That is accounting profit. After the owner's implicit costs of Tk 887,000, the shop's economic profit is -Tk 137,000.

Mistake: "Interest is only an unfair charge on borrowers." Interest pays the lender for waiting and for risk, and it rations scarce funds among borrowers who want them. Whether a particular rate is fair is a separate question.

Mistake: "Normal profit is a form of profit left for the owner." Normal profit is part of costs, because it covers the owner's implicit costs. Only economic profit above zero is a surplus above all costs.

Mistake: "Profit has one cause." Risk, the owner's contribution, innovation and monopoly power each explain some profits and fail for others.

* * *

IN HISTORY

In 1976, Muhammad Yunus, a Bangladeshi economist then teaching at Chittagong University, began to lend small sums to poor villagers in Jobra, a village near the university. Banks at that time refused to lend to people who owned no property. Yunus found that the borrowers repaid, and he went on to found the Grameen Bank in 1983.

The Grameen Bank grew into a large lender to poor households in Bangladesh, and in 2006 Yunus and the bank shared the Nobel Peace Prize. The story links to this chapter in two ways. Loans to people without property carry more risk and cost more to administer, which usually raises the rate. Yet repayment turned out better than lenders had expected, so the chance of loss was smaller than they feared.

* * *

11.3 Marginal cost per page for each step of five pages: Tk 20, 16, 12, 16, 22 and 26. For example, $(300 - 200) \div 5 = 20$.

11.4 $ATC = TC \div Q$: Tk 60.00, 38.00, 29.33, 26.00, 25.20 and 25.33 at 5 to 30 pages. It is lowest at 25 pages, at Tk 25.20.

11.5 Total cost for one day at 20 pages is Tk 520, which includes the fixed cost. For 6 days, $520 \times 6 = \text{Tk } 3,120$.

11.6 Total cost at 20 pages rises by Tk 60, from 520 to 580. Marginal cost does not change, because fixed cost cancels in the subtraction. Average total cost at 20 pages is $580 \div 20 = \text{Tk } 29$.

11.7 Forgone interest = $0.08 \times 3,000,000 = \text{Tk } 240,000$. Implicit costs = $360,000 + 240,000 = \text{Tk } 600,000$. Result = $500,000 - 600,000 = -\text{Tk } 100,000$.

CHAPTER XII · THE LEAST-COST COMBINATION OF INPUTS

12.p1 $4,800 / 600 = 8$ workers.

12.t1 The firm moves toward machines: fewer workers and more machines. At the new wage the least-cost mix is (3, 6) at Tk 5,100. Keeping (4, 4) would cost Tk 5,200, a rise of $4 \times 300 = \text{Tk } 1,200$. Changing the mix limits the rise to Tk 1,100.

12.t2 No. With 6 machines fixed, the mix for 100 books is (3, 6), which costs Tk 4,200. That is Tk 200 above the least cost of Tk 4,000 at (4, 4). Only workers can be changed in the short run.

12.1 An isoquant joins all mixes of two inputs that give the same output. An isocost line joins all mixes that cost the same total amount. Output can be counted in units, but the satisfaction shown by an indifference curve cannot be counted.

12.2 All workers: $4,800 / 600 = 8$ workers. All machines: $4,800 / 400 = 12$ machines. The slope is the price ratio $600 / 400 = 1.5$.

12.3 Costs: (1, 8) = $30 + 80 = \text{Tk } 110$; (2, 4) = $60 + 40 = \text{Tk } 100$; (4, 2) = $120 + 20 = \text{Tk } 140$; (8, 1) = $240 + 10 = \text{Tk } 250$. The least-cost combination is (2, 4) at Tk 100.

12.4 Costs: (2, 10) = $1,200 + 3,000 = \text{Tk } 4,200$; (3, 6) = $1,800 + 1,800 = \text{Tk } 3,600$; (4, 4) = $2,400 + 1,200 = \text{Tk } 3,600$; (5, 3) = $3,000 + 900 = \text{Tk } 3,900$; (6, 2) = $3,600 + 600 = \text{Tk } 4,200$. Two mixes tie at Tk 3,600. The MRTS between them is 2, which equals the new price ratio $600 / 300 = 2$.

12.5 At (4, 4): $2,400 / 4,000 = 60\%$. At (3, 6) with the wage at Tk 900: $2,700 / 5,100 = 52.9\%$. The share falls because the firm replaces workers with machines.

12.6 Workers: $30 / 600 = 0.05$ books per taka. Machines: $10 / 400 = 0.025$ books per taka. Workers add more per taka, so the firm should hire more workers and rent fewer machines.

CHAPTER XIII · MARKET STRUCTURE AND PERFECT COMPETITION

13.p1 Its sales fall to nothing. Buyers can obtain an identical good at Tk 40 from other firms and know that price, so no one pays Tk 42.

13.t1 Other firms see the profit and, since entry is free, they begin to sell the good. Industry supply shifts to the right, the price falls, and the fall stops when no profit above normal remains.

CHAPTER VIII · SUPPLY AND THE SUPPLY CURVE

8.p1 50 notebooks. A higher price (Tk 50 against Tk 30) brings a larger quantity supplied, which is the law of supply.

8.p2 A shift to the right, because a subsidy lowers the seller's expense, so more notebooks are offered at every price. The price of notebooks itself did not change.

8.t1 At a high rate of pay the worker can earn the income needed in fewer hours, and may value rest more. The curve rises, reaches a peak and bends backward.

8.t2 For the day, the curve is a vertical line at 200 fish, since more cannot be had at any price. Over a week more fish can be brought, so the quantity supplied can rise with the price and the curve slopes upward.

8.1 The points are (20, Tk 20), (30, Tk 30), (40, Tk 40), (50, Tk 50), (60, Tk 60), with quantity on the horizontal axis. The line slopes upward: each Tk 10 rise in price adds 10 notebooks, so a higher price goes with a larger quantity supplied.

8.2 The new curve lies 10 notebooks to the left of the old one at every price. An increase of 15 on the old curve gives $Q_s = P + 15$, so the quantity at Tk 40 is 55 ($40 + 15$).

8.3 (a) movement; (b) shift, to the right; (c) shift, to the right; (d) movement.

8.4 At Tk 55, $Q_s = 55$ notebooks. For $Q_s = 25$, $P = \text{Tk } 25$. With $Q_s = P - 10$, at Tk 55 the quantity is $55 - 10 = 45$.

8.5 When the price of a good rises, the quantity supplied rises, if other conditions stay the same. Costs of inputs, technology, taxes and subsidies, or the number of sellers must stay fixed.

8.6 A backward-bending curve, such as hours of work offered at a very high rate of pay; and a fixed supply, such as the fish caught for one market day.

8.7 Supply rises, so the curve shifts to the right. The number of sellers has changed, and that is a determinant of supply.

CHAPTER IX · ELASTICITY OF SUPPLY AND MARKET EQUILIBRIUM

9.p1 Buyers want $80 - 40 = 40$ notebooks, and firms offer 40 notebooks. The two quantities are equal.

9.p2 Both rise. The shift creates excess demand at the old price, the price climbs, and firms offer more along the supply curve.

9.t1 At Tk 30, buyers want 50 notebooks and firms offer 30, so excess demand is 20. Buyers who obtain a notebook gain a lower price; the 20 who cannot buy, and the firms who cannot charge more, lose.

9.t2 She describes time. This month supply is inelastic because the crop cannot grow faster; next year it is more elastic because she can plant more.

9.1 $\Delta Q = 20$, average $Q = 100$, so 0.2. $\Delta P = 10$, average $P = 25$, so 0.4. Elasticity = $0.2 \div 0.4 = 0.5$. Supply is inelastic.

9.2 $\Delta Q = 100$, average $Q = 450$, so 0.222. $\Delta P = 30$, average $P = 115$

quantity demanded; the whole list is demand. For the notebook: 40 at Tk 40, 50 at Tk 30, and so on. (*Chapter IV*)

Demand curve. A line on a graph showing the quantity demanded at each price. Price is on the vertical axis and quantity on the horizontal axis. A point on the line means that at this price buyers choose this quantity. Usually the line slopes downward. (*Chapter IV*)

Demand schedule. A table showing the quantity demanded at each price. Each row pairs one price with one quantity. The shop's table lists Tk 60 with 20 notebooks, Tk 50 with 30, and continues down to Tk 20 with 60. (*Chapter IV*)

Economic law. A statement of how people usually behave, true when other conditions stay the same. It describes a tendency, not a certainty, because people choose. When the price of a good rises, consumers usually buy less of it. (*Chapter I*)

Economic profit. Revenue minus all costs, both explicit and implicit, including normal profit. It is accounting profit less the value of what the owner's time and money could earn elsewhere. A positive result is supernormal profit, and a negative result is below normal. (*Chapter XVI*)

Economic rent. The payment to any factor above the least needed to keep it in use. For land as a whole the least needed is nothing, so the whole payment is economic rent. Example: a singer paid Tk 100,000 who would sing for Tk 40,000 earns Tk 60,000. (*Chapter XV*)

Economics. The study of how people, firms and governments choose to use limited resources to satisfy wants. It exists because wants are many and means are few. A student with Tk 100 who must choose among wants worth Tk 150 meets the problem on a small scale. (*Chapter I*)

Elastic demand. Demand for which the percentage change in quantity is larger than the percentage change in price, so price elasticity of demand is above 1. A price cut raises total revenue. Example: the notebook between Tk 50 and Tk 40, at 1.29. (*Chapter V*)

Elasticity of supply. The percentage change in quantity supplied divided by the percentage change in price. It tells us how strongly firms respond to a price change. If a 10% rise in price raises quantity supplied by 5%, the elasticity of supply is 0.5. (*Chapter IX*)

Equilibrium of the firm. The output at which a firm earns the largest possible profit and has no wish to change. A price taker finds it where marginal cost equals the price. For the binding workshop at Tk 40, it is 50 units. (*Chapter XIII*)

Equilibrium price. The price at which the quantity buyers want equals the quantity sellers offer. At this price neither side has a reason to change its plans, so the price stays still. In the notebook market it is Tk 40. (*Chapter IX*)

Exceptional supply curve. A supply curve that does not slope upward, such as one that bends backward or stays fixed. A worker who offers fewer hours at very high pay gives a backward-bending

4.2 The new curve lies 5 notebooks to the left of the old one at every price. A rise of 10 on the old curve gives $Q = 90 - P$, so the quantity at Tk 40 is 50 ($40 + 10$).

4.3 (a) movement; (b) shift, to the right, since pens and notebooks are used together; (c) shift, to the left; (d) movement.

4.4 At Tk 35, $Q = 80 - 35 = 45$ notebooks. For $Q = 55$, $P = 80 - 55 = \text{Tk } 25$.

4.5 Substitute goods: tea and coffee; a price rise for tea raises demand for coffee. Complementary goods: pens and notebooks; a price rise for pens lowers demand for notebooks.

4.6 A Giffen good is a rare, very basic good for which quantity demanded rises when its price rises. It is not the usual case because for almost every good a higher price lowers the quantity demanded.

4.7 Demand rises, so the curve shifts to the right. The number of buyers has changed, and that is a determinant of demand.

CHAPTER V · ELASTICITY OF DEMAND

5.p1 $\% \Delta P = 4 \div 10 \times 100 = +40\%$; $\% \Delta Q = -10 \div 25 \times 100 = -40\%$; $E = 40 \div 40 = 1$, so demand is unit elastic.

5.t1 Quantity demanded falls by about $0.3 \times 10\% = 3\%$. Demand is inelastic, so total revenue rises, because the 10% higher price outweighs the 3% fall in quantity.

5.t2 With time, consumers find substitutes and change habits, so demand becomes more elastic. A large rise may also lose customers who watch the price closely, so the gain in revenue may not last.

5.1 It is the percentage change in quantity demanded divided by the percentage change in price. Percentages remove units and starting levels, so goods of different size can be compared.

5.2 The base is the average of the old and new values, which is the same whichever is called old. The percentage and so the elasticity do not depend on the direction.

5.3 $\% \Delta Q = -20 \div 50 \times 100 = -40\%$; $\% \Delta P = 10 \div 25 \times 100 = +40\%$; $E = 1$. Demand is unit elastic.

5.4 Quantity rises from 40 to 50: $\% \Delta Q = 10 \div 45 = 22.22\%$; price falls from 40 to 30: $\% \Delta P = 10 \div 35 = 28.57\%$; $E = 0.78$, inelastic. Total revenue falls from Tk 1,600 to Tk 1,500, as the test predicts.

5.5 Income elasticity = $10\% \div 20\% = 0.5$.

5.6 Cross elasticity = $-3\% \div 5\% = -0.6$. The negative sign shows that tea and sugar are used together.

5.7 A brand of tea has many close substitutes, namely other brands, so a consumer can switch when its price rises. Tea in general has fewer substitutes.

CHAPTER VI · CONSUMPTION AND CONSUMERS' SURPLUS

6.p1 No. The price of Tk 12 is above the willingness to pay of Tk 9, so the consumer does not buy the pen.

6.t1 Each old buyer keeps Tk 10 more, because the price fell by Tk 10. The new buyers are those with a willingness to pay of Tk 10 or more. The total surplus is the area under the demand curve and above the price of Tk 10. This is a triangle with a base of 10 (from 0 to 10 on the quantity axis) and a height of 10 (from 10 to 20 on the price axis). The area is $\frac{1}{2} \times 10 \times 10 = 50$. The total surplus is Tk 50.

and machines all give 100 books a day, so they lie on one isoquant. (*Chapter XII*)

Law of demand. When the price of a good rises, the quantity demanded falls, if other conditions stay the same. The same law says a fall in price raises the quantity demanded. It is drawn as a downward-sloping line. (*Chapter IV*)

Law of diminishing marginal utility. As a person consumes more units of a good, the satisfaction added by each extra unit falls. It holds when units are alike, tastes are steady and consumption has no long gaps. Marginal utilities of 10, 8, 5, 2 and 0 show the law. (*Chapter III*)

Law of diminishing returns. Adding more of one input to fixed amounts of others will, after a point, add less and less to output. It belongs to the short run and does not say that total output falls. Example: each worker beyond the second in the workshop. (*Chapter X*)

Law of supply. When the price of a good rises, the quantity supplied rises, if other conditions stay the same. The same law says a fall in price lowers the quantity supplied. It is drawn as an upward-sloping line. (*Chapter VIII*)

Least-cost combination. The mix of inputs that produces a given output at the lowest total cost. It lies where the lowest isocost line touches the isoquant. In the case, 4 workers and 4 machines make 100 books for Tk 4,000, less than any other mix on the isoquant. (*Chapter XII*)

Liquidity preference. The wish of people to hold money in cash form instead of lending it, because cash can be spent at once. Interest is the reward that persuades people to give up cash. Keeping money for emergencies is an example. (*Chapter XVI*)

Loanable funds. The money that savers offer to lend and that borrowers demand to borrow at each rate of interest. Borrowers want more when the rate is low, and savers offer more when it is high. The rate where the two amounts are equal is the equilibrium rate. (*Chapter XVI*)

Long run. A period long enough for all inputs to be changed. Its length differs between firms. Example: given years, a baker can build a new bakery and buy more ovens. (*Chapter X*)

Macroeconomics. The branch of economics that studies the whole economy, including output, prices and employment. Its unit is a total or an average for a country. A question about the price level of a whole country belongs to macroeconomics. (*Chapter I*)

Marginal cost. The extra cost of producing one more unit of output, found by dividing the change in total cost by the change in output. Fixed cost does not affect it. If total cost rises by Tk 220 for 10 more books, marginal cost is Tk 22 a book. (*Chapter XI*)

Marginal product. The extra output from adding one more unit of an input, others held fixed. It equals the change in total product divided by the change in the input. Example: the fourth worker adds $44 - 36 = 8$. (*Chapter X*)

ANSWERS TO QUESTIONS

CHAPTER I · WHAT ECONOMICS IS

1.p1 (a) Positive: the shop's staff records can confirm the number 7. (b) Normative: whether the shop ought to employ an eighth person depends on values and judgement, and facts alone cannot settle it.

1.t1 The claim ignores that another condition changed in the same month, since free notebooks reduce the need to buy. *Ceteris paribus* requires all other things to stay the same, and here they did not. We cannot tell how much of the fall came from the price.

1.t2 Economic laws describe how people usually behave, so they state tendencies and not certainties. Exactness is not the test of usefulness; a usual pattern can still guide a seller or a government. Used with *ceteris paribus*, such a law separates one cause from many.

1.1 (a) Positive: staff records can test it. (b) Normative: it says what ought to be. (c) Positive: the accounts can confirm the figure. (d) Normative: should signals a value judgement.

1.2 (a) Micro: one firm. (b) Macro: the general price level. (c) Micro: one consumer. (d) Macro: a total for the whole economy.

1.3 Her wants cost Tk 60 + Tk 60 + Tk 30 = Tk 150, but she has only Tk 100. Wants exceed means by Tk 50, so she must choose and give up something.

1.4 Consumption, production, exchange and distribution. Examples: a reader takes a book home (consumption); a printer makes the book (production); the shop buys it from the printer (exchange); the shop's income pays staff and owner (distribution).

1.5 *Ceteris paribus* means all other things stay the same. An economic law states a usual tendency of people, while a law of nature, such as the fall of a stone, describes things that do not choose and holds exactly.

CHAPTER II · SCARCITY, CHOICE AND OPPORTUNITY COST

2.p1 The best alternative given up, which is the Tk 300 earned. Sleeping late is a lesser alternative and is not added.

2.t1 The cost is the Tk 120 that could have been earned, the best alternative. For a student with no job offer, or with a different alternative such as a tutoring fee, the cost would differ.

2.t2 The maximum is 1,000 books or 500 parcels. 600 books needs 30 hours and 300 parcels needs 30 hours, so 60 hours are needed; it is still not attainable.

2.1 A want is something a person would like to have, whatever the price. One want may be met, but others appear in its place, so the whole list is without limit.

2.2 Scarcity is the condition that resources are limited while wants are unlimited, so choices must be made. A rich person has limited time; a poor person has limited money.

2.3 The Tk 400 from working, as the best alternative. Resting is a

all alternatives added. Example: Rafiq's extra parcel gives up 2 books. (*Chapter II*)

Perfect competition. A market with very many buyers and sellers, identical goods, free entry and full information. No real market meets all four conditions, so it is a model. A grain exchange comes near it, because wheat of one grade is the same whoever grew it. (*Chapter XIII*)

Positive economics. Statements about what is, which can be tested against facts. A positive statement may be true or false, but records or measurement can settle which. The shop sold 41,000 copies in 2025 is a positive statement. (*Chapter I*)

Price elasticity of demand. The percentage change in quantity demanded divided by the percentage change in price, each found by the midpoint method. A result of 2 means quantity changes twice as fast as price. The size of the result is reported, without its sign. (*Chapter V*)

Price maker. A firm that can choose its price because it faces the whole demand of the market, as a monopolist does. It cannot choose both price and quantity, since the demand curve fixes one once the other is chosen. Example: the seller in the case file, who chooses Tk 50. (*Chapter XIV*)

Price taker. A firm that must accept the market price because its own sales are too small to affect it. If the market price is Tk 40, such a firm sells any quantity at Tk 40 and nothing at a higher price, so it chooses only its output. (*Chapter XIII*)

Product differentiation. Making a good different from rivals' goods in quality, style or brand, to build loyal buyers. The differences may be real or only believed. Example: a stall that serves tea in clean glasses with a friendly greeting. (*Chapter XIV*)

Production function. A rule showing the largest output that can be made from each combination of inputs. It records what is technically possible with the best known methods. Example: 1 oven and 2 helpers can make at most 100 loaves a day. (*Chapter X*)

Production possibility curve. A line showing the largest combinations of two goods that can be produced with all resources used fully. Points inside waste resources; points outside cannot be reached. Example: Rafiq's 800 books or 400 parcels. (*Chapter II*)

Quasi-rent. A temporary surplus earned by a man-made asset while its supply is fixed in the short run. It disappears when new assets are built. Example: printing presses when demand for books suddenly rises. (*Chapter XV*)

Real wage. A wage measured by the goods and services it can buy. It equals the nominal wage divided by the price index over 100. Example: $35,000 \div 1.12 = \text{Tk } 31,250$. (*Chapter XV*)

Rent. The payment to the owner of land for its use, in everyday talk and in economics. In this book the word is kept for land. Example: Tk 60,000 a month paid for a shop site. (*Chapter XV*)

Returns to scale. The change in output when all inputs are increased in the same proportion. It may be increasing, constant or

Willingness to pay, 53–58, 65